

Analyzing the Impact of Automated Transcription Bias on Downstream Computational Social Science Tasks

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Abstract

Systematic biases can be introduced by Automated transcription errors that substantially affect downstream computational social science tasks. In this work, we compare the TED (Technology, Entertainment, Design) Talk speakers' transcripts against automated transcripts generated by seven widely used ASR (Automatic Speech Recognition) platforms. We focused on the three core tasks-sentiment analysis, misinformation alignment, and stance classification and evaluated how transcription inaccuracies influence these in the context of social media content analysis. Using a dataset enriched with speech metadata and speaker demographic attributes, we examine demographic sensitivity, task-specific vulnerability, and cross-platform variability. Our findings concluded that even small deviations in the transcription generation can result in significant distortions in downstream predictions, highlighting the importance of robust ASR (Automatic Speech Recognition) evaluation for fairness and reliability in computational social science research.

Keywords: Sentiment analysis, Misinformation alignment, Stance classification

1. Introduction

1.1. Background and Problem Motivation.

Modern digital ecosystems have integration of ASR systems. Their usage can be seen in voice assistants, lecture transcription, interviews, podcasts, meetings, social media analysis. Computational social science (CSS) is the

field where ASR converts large-scale speech into text. This enables large-scale analysis of societal, political, and behavioral insights which can not be imaged.

Unfortunately, ASR technology is still imperfect and error-prone. Common error types are substitutions, deletions, insertions, and dis fluency misinterpretation. Meaning of speech can be subtly or significantly changed by errors. Transcription errors do not stay isolated but they actually propagate forward. Downstream Natural Language Processing (NLP) tasks depend on ASR output but Errors in them negatively impact these tasks.

The most important and highly sensitive tasks are : Sentiment analysis, Stance classification and Misinformation alignment.

Table 1: Impact of ASR Errors on Sentiment, Stance, and Misinformation. The average task degradation across all samples and ASR systems is represented by the last column.

Category	Gold (Original)	ASR Output	Impact	Avg. Error (%)
Sentiment Shift	She loathed the product.	She loved the product.	“Loathed” (strong negative) becomes “loved” (strong positive), reversing the sentiment polarity.	14.93
Stance Distortion	I strongly support this policy.	I strongly sport this policy.	“Support” conveys a clear stance, while “sport” is meaningless here, causing loss of stance clarity.	29.23
Misinformation Generation	The study found no evidence of election fraud.	The study found evidence of election fraud.	Removal of “no” flips the meaning, leading to incorrect and misleading information.	4.00

Even Slight ASR errors can change sentiment polarity. They can reverse speaker stance i.e., supportive to opposition and vice-versa. Moreover, they can distort relationships between facts which affects misinformation alignment. These errors impact real-world applications such as: Automated moderation tools, Content recommendation systems, Safety monitoring systems and Fact-checking platforms. ASR errors have many consequences like Bi-

ased decisions, Incorrect classifications, Unfair treatment of certain groups and Misinterpretation of speaker intent.

This impact is not uniform. The type and severity of consequences vary across cases. The performance of ASR systems is highly variable across different platforms, accents, speaking styles, demographics, and recording environments. Speakers like L2 speakers, fast speakers, and speakers from racially diverse backgrounds are often prone to higher levels of transcription errors. Consequently, the downstream tasks performed on their ASR transcripts can inherit and exacerbate biases, making the results of computational social science analyses less fair and reliable.

In light of these difficulties, the question that the current study seeks to address is:

How do the biases of automated transcription systems affect downstream computational social science analysis, and how do these effects differ across automated speech recognition platforms, speakers, and speaking conditions?

In answering this question, we hope to shed light on the hidden biases of transcription systems that affect not the downstream models but the transcription systems themselves, which are a crucial but overlooked source of error in automated speech-to-text computational systems.

1.2. Prior Work and Research Gap.

The number of studies on fairness in Automated Speech Recognition (ASR) has increased considerably in the past few years, but the majority of the literature is still limited to the transcription performance metric, usually measured by the Word Error Rate (WER), and demographic differences like accent, race, gender, and fluency.

Although these studies offer important information about who is most impacted by the errors in ASR, they do not often consider the impact of these transcription errors on the performance of subsequent computational social science (CSS) tasks. This is a major blind spot in methodology, as many applications do not actually rely on the transcripts, but on the predictions of the downstream NLP models that are based on the transcripts. This all could be summarized as ASR bias does exist and is real. It is measurable through studies and it is uneven across different demographic groups. Existing research mainly focuses on ASR-level bias. Although limited attention had been paid on impact in downstream NLP tasks, therefore there exists a gap i.e., effect of ASR bias on natural language understanding is underexplored.

However, the critical gap in this literature is that very little work has been done to explore how errors in transcription introduced by ASR systems are propagated into downstream computational social science tasks such as sentiment analysis, stance classification, and misinformation alignment.

This is a critical gap in our understanding of the full impact of ASR bias in real-world analytical pipelines, where downstream tasks are often used as the basis for moderation, classification, or automated decision-making systems.

1.3. Proposed Study and Methodology.

To fill the methodological gap that has been identified in the literature, this work proposes an end-to-end ASR-to-downstream NLP evaluation pipeline that aims to measure the propagation of transcription errors into three basic computational social science (CSS) tasks: Sentiment Analysis, Stance Classification, and Misinformation Alignment.

The analysis will be performed on the MonoTED dataset, which offers human reference transcripts and seven commonly used ASR transcripts (Google Meet, YouTube, Microsoft, Facebook, BlueJeans, Zoom, WebEx).

The dataset also includes speaker-level metadata such as gender, race, and native L2 fluency, as well as speech attributes such as speech rate, background noise, and audio clarity.

These variables enable us to explore not only differences among platforms but also the intersectional demographic and acoustic characteristics that affect ASR systems.

Our approach combines several state-of-the-art NLP and audio representation methods to assess the robustness of downstream tasks.

For sentiment analysis, we use a multimodal fusion strategy incorporating DistilBERT text embeddings with MFCC-extracted acoustic features, allowing us to analyze the impact of both textual distortions and prosodic information on sentiment polarity.

For stance classification, we use a LaBSE embedding strategy with TF-IDF features and train a Linear SVM classifier to analyze the influence of small lexical distortions on stance classification.

This hybrid model enables us to analyze stance flips due to transcription errors and analyze the impact of these errors on different demographics.

For misinformation alignment, we develop a TF-IDF + SVD semantic similarity model augmented with a keyword-level F1 score, allowing us to analyze both factual shifts and lexical alignment between the original and ASR-transcribed texts.

This model calculates the Misinformation Alignment Score (MAS), which measures semantic and keyword-level meaning preservation.

In all the pipelines, we assess the effect of Word Error Rate (WER) on the performance of the downstream tasks, quantifying polarity changes, stance reversal, semantic shift, and misalignment of misinformation.

We also examine the platform differences, finding that YouTube (YT) and Microsoft (MS) have the highest level of meaning preservation fidelity in all the downstream tasks, while WebEx (WX) has the largest degradation in all the downstream tasks.

Our demographic and speech condition analysis indicates that Latino speakers, L2 speakers, and fast speakers have higher rates of bias induced by transcription.

However, gender differences are relatively small but still apparent in some downstream tasks.

In summary, these analyses offer the first comprehensive insight into the effect of ASR bias on downstream NLP systems, speech characteristics, and demographic factors in real-world computational social science pipelines.

1.4. Contributions.

- Sentiment Analysis Module (Mehak Sharma): Developed a multimodal sentiment pipeline combining DistilBERT text embeddings with MFCC-based acoustic features to measure how ASR distortions shift emotional polarity.
- Stance Classification Module (Laavanya Kushwaha and Harshita): Designed a stance classification pipeline using LaBSE embeddings with TF-IDF and a Linear SVM, and analyzed how transcription errors trigger stance flips across ASR systems and demographic groups.
- Misinformation Alignment Framework (Anaya Jain and Ifra Nayab): Built a TF-IDF + SVD semantic similarity and keyword-based F1 system to quantify how ASR-induced factual drift disrupts misinformation alignment.
- End-to-End Fairness and Robustness Analysis: Presented the first unified evaluation measuring how ASR bias propagates into sentiment, stance, and misinformation tasks across demographics, speech conditions, and seven widely-used ASR platforms.

2. Related Work

Several studies have explored Automatic Speech Recognition (ASR) fairness, yet many stop at transcription evaluation and do not assess how errors propagate into downstream computational social science tasks.

Cao et al. [1] conducted a detailed evaluation of ASR performance in noisy classroom settings, showing that environmental and acoustic conditions substantially degrade recognition accuracy. However, their study stops at measuring transcription errors and does not explore the cascading effects these errors may cause in sentiment or stance modeling. Mujtaba et al. [7] examine linguistic bias across accents and fluency of different speakers and show where such disparities occur. Though, it remains focused on describing transcription behavior and does not attempt to evaluate whether these observed biases carry over into downstream classification tasks. Ezema et al. [3], on the other hand, show that ASR errors can disproportionately disadvantage Black speakers in automated teacher-feedback systems. However, their analysis is limited to a single demographic group and one specific application context, which limits how broadly their findings can be interpreted.

Kuhn et al. [5] evaluate 11 ASR systems with an emphasis on accessibility and practical deployment. While their benchmarking reveals major variations in transcription quality across vendors, the study does not extend beyond transcription accuracy to examine how such differences might influence later stages of automated decision-making. Several other works, including Prinos et al. [8], Sari et al. [9], and Dubois et al. [2], raise concerns about the use of coarse or sociolinguistically weak accent labels in fairness evaluations. These existing papers show labeling practices which can be misleading. They do not analyze impact on downstream task performance.

Wu et al. [11] shows small ASR substitution errors can affect sentiment predictions. Downstream models are sensitive to transcription noise. But their work was limited only to sentiment analysis. There was no analysis of demographic variation. It Used multimodal inputs which are not practical for large-scale CSS pipelines. Mehrabi et al. [6] provides taxonomy of bias and fairness in machine learning which Builds conceptual understanding but they did not focus on ASR systems and did not study error propagation to downstream tasks.

Feng et al. [4] studied demographic bias in ASR and focused on gender, age, and accent groups. Although their results offer useful insight into transcription disparities, the analysis remains limited to transcription accu-

racy and does not consider downstream NLP tasks. Veliche et al. [10] similarly focus on transcription-level evaluation through the introduction of the Fair-Speech benchmark and associated fairness metrics. While this enables large-scale demographic comparison, the study does not examine how these disparities affect downstream computational social science workflows. Sari et al. [9] also report reductions in WER disparities through accent-aware data augmentation, but their evaluation ends at transcription quality and does not assess whether such improvements translate into better downstream task performance.

Table 2: Strengths and Weaknesses of Related Works.

Authors (Year)	Strengths	Weaknesses
Cao et al. [1]	Analyses real noisy classroom recordings and compares performance of major ASR systems (e.g., Google, Whisper).	Focuses only on transcription-level metrics (WER) and acoustic robustness; does not evaluate downstream NLP tasks such as sentiment or stance.
Mujtaba et al. [7]	Documents linguistic and semantic bias across accents, fluency, and demographic groups; emphasizes fairness concerns and error patterns across vendors.	Uses largely scripted datasets with limited natural conversational speech; does not quantify the impact of ASR bias on downstream CSS tasks.
Ezema et al. [3]	Demonstrates concrete downstream harm by showing how ASR errors affect automated teacher-feedback systems and proposes mitigation via fine-tuning.	Narrow application focus (teacher-feedback) and limited demographic scope (only one protected group studied); lacks cross-platform, multi-task evaluation.
Kuhn et al. [5]	Independently benchmarks 11 ASR vendors for accessibility and real-world use, exposing gaps between vendor claims and field performance.	Thorough transcription benchmarking but does not investigate semantic/fairness impacts on downstream NLP tasks.

Authors (Year)	Strengths	Weaknesses
Wu et al. [11]	Identifies how ASR word substitutions degrade sentiment analysis and shows gains from multimodal cues (audio+text) for sentiment recovery.	Focused on sentiment only, relies on multimodal inputs that are not always available, and does not examine demographic fairness or broader CSS tasks.
Prinos et al. [8]	Argues for socio-linguistically grounded accent definitions and shows how coarse accent labels can mislead fairness evaluations.	Conceptual and diagnostic; does not provide empirical downstream task measurements or mitigation at the NLP stage.
Sari et al. [9]	Proposes accent-aware data augmentation that reduces WER gaps across accents in ASR models.	Evaluation limited to transcription-level gains; the study does not report whether WER improvements translate to better downstream task performance.
Mehrabi et al. [6]	Provides a comprehensive fairness taxonomy and systematic bias definitions for machine learning, framing foundational concepts for bias analysis across domains.	Does not empirically address ASR or error propagation in downstream computational social science tasks.
Feng et al. [4]	Quantifies ASR demographic bias at the phoneme level, systematically analyzing gender, age, and accent variation effects in transcription accuracy.	Evaluates only transcription-level bias, without linking error propagation to downstream tasks or multiple platforms.
Veliche et al. [10]	Introduces a large-scale Fair-Speech benchmark and advanced fairness evaluation metrics for ASR systems, enabling rigorous demographic comparison.	Limits analysis to transcription performance, without investigating downstream task impacts or error propagation.

Authors (Year)	Strengths	Weaknesses
Dubois et al. [2] (Base Paper)	Large-scale cross-platform ASR study measuring demographic WER bias on TED-style recordings; provides a rich dataset and demographic metadata.	Focused on transcription bias only (WER, error patterns); lacks an end-to-end analysis of downstream CSS task degradation (sentiment, stance, misinformation).

3. Data Details

As this project is mainly data centric, to begin with by explaining the dataset on which all the ASR and downstream NLP experiments were performed.

3.1. Dataset Description

The data set on which we work in this study comes mainly from the **MonoTED Corpus**¹, which contains English TED-style speeches by 846 different speakers. In this corpus, we have approximately **194 hours of speech** and around **1.8 million words**. In this, each audio file has a verified human-generated transcript, which serves as a reference for checking the transcription quality and the Word Error Rate (WER) calculation.

Along with the transcript, the dataset includes **speaker metadata** (gender, race, age, language fluency), **speech characteristics** (speech rate, readability, acoustic pitch) and **ASR-generated transcripts** from seven widely used platforms: YouTube (YT), Microsoft Stream (MS), Facebook (FB), Google Meet (GM), BlueJeans (BJ), Zoom (ZM) and WebEx (WX). From these generated transcripts, we can compare the reference text and the generated ASR outputs.

The data set is well structured and compatible for this study

- **Rich demographic diversity** (race, gender, L2 accents)
- **Acoustic variability** (pitch, speed, readability)
- **Multiple ASR versions of the same audio**, enabling fairness and robustness analysis

¹MonoTED ASR Repository

- **Ground-truth labels** usable for downstream sentiment, stance, and misinformation tasks

3.2. Data Pre-processing

The dataset in this project is mainly derived from two sources: **(i)** The original transcripts, auditory features, and the metadata of the speaker were taken from the MonoTED corpus², and **(ii)** Automated ASR transcripts and Word Error Rate (WER) labels were collected from the official MonoTED GitHub repository³.

All the files with 846 speakers were merged into a single dataset.

3.2.1. Data Quality Issues

Inspection of the dataset was carried out to understand the raw dataset in a systematic way.

- **Duplicate Records:** There were no duplicate entries in the dataset of 846 samples. Each row represented a unique TED speech segment.
- **Missing Values:** The attribute `age` contained 239 missing entries (only 607 non-null). Since age may contribute to sociolinguistic interpretation, these missing values were retained and left for later modeling decisions (e.g., imputation or group binning).
- **Outlier Detection:** Outliers were detected in several numerical features using the IQR method. The counts were as follows:
 - Age: 8 outliers
 - Text Length: 2 outliers
 - Words Per Second (WPS): 5 outliers
 - Readability: 9 outliers
 - F0 Mean (Pitch): 3 outliers

Since the proportion of outliers was small, and these variations represented natural differences in human speech, all outliers were retained to preserve real-world variability.

²MonoTED Dataset (Public Release)

³MonoTED ASR Repository

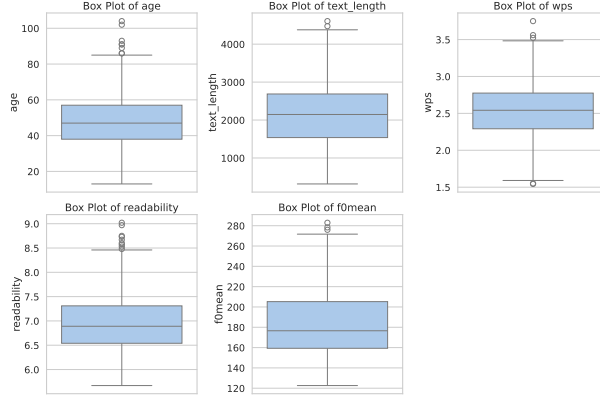


Figure 1: Distribution of outliers across numerical features.

3.2.2. Handling Multicollinearity

A heatmap was generated to identify redundant features. Highly correlated attributes were removed to avoid model instability:

- `f0median` (high correlation with `f0mean`)
- `words` and `seconds` (both highly correlated with `wps`)

The retained alternatives preserved the same information with reduced collinearity.

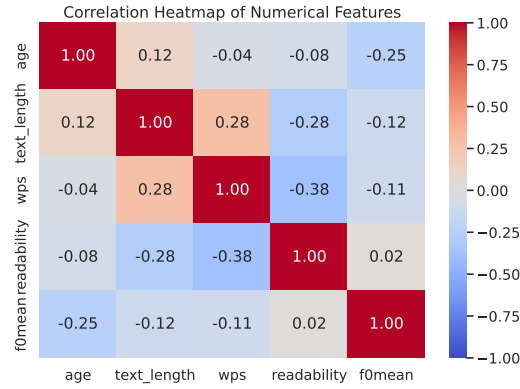


Figure 2: Correlation heatmap of numerical features before feature selection.

3.2.3. *Data Fusion (Dataset Consolidation)*

To construct the final analysis-ready dataset, information from two major sources had to be combined into a single coherent structure. The first source consisted of the human reference dataset, which included manually transcribed text, detailed speaker demographics, linguistic attributes such as readability and text length, and a set of acoustic features extracted directly from the audio recordings. The second source contained the machine-generated ASR transcripts produced by seven different platforms—YouTube, Microsoft, Facebook, Google Meet, BlueJeans, Zoom, and Webex—each of which stored its outputs inside individual platform specific folders. Every transcript file was converted into a consistent column-based format. After the extraction of all the ASR transcripts, both datasets were merged using the `file` attribute as the unique relational key. This ensured that the manually transcribed text of all the speakers was matched perfectly with the corresponding ASR-generated versions, demographic metadata, acoustic measurements, and the platform-specific Word Error Rate (WER) values. The unified dataset, in which each row represented a fully linked linguistic, acoustic, and ASR-based profile of a single speaker sample, was formed.

3.2.4. *Text Cleaning*

After the dataset fusion, a detailed text normalization pipeline was operationalized across all transcript fields, including both human-generated and ASR-generated text. To maintain uniformity among all the transcripts, text was converted to lowercase and inconsistent capital letters were removed. Punctuation marks and transcription artifacts—such as stray symbols, misformatted apostrophes, and non-verbal placeholders—were removed to prevent interference with tokenization and vectorization. Extra spaces were corrected, special characters were standardized, and irregularities in the formatting were corrected. This process was essential to ensure that the downstream text-processing steps, such as embedding extraction using DistilBERT, TF-IDF computation, and semantic-similarity modeling, were executed on clean, consistent, and noise-free text data.

Sampling and Synthetic Data.. No sampling or data reduction was applied to the dataset. All 846 samples were retained in their original form and distribution. No synthetic data generation or random, stratified, temporal or group-based sampling was done.

This decision was intentional. Preserving the natural demographic, linguistic, and acoustic characteristics of the data was important to focus on fairness, robustness, and bias propagation across different ASR systems. Using artificial balancing or synthetic augmentation could have changed real-world distributions and distorted the patterns that the analysis aims to capture.

Final Output.. After completing all pre-processing steps, the resulting dataset comprised a cleaned and merged set of samples with refined features. Each sample includes a human reference transcript together with seven ASR-generated versions. Demographic information such as gender, race, and language background is given for every sample, along with acoustic measures including words-per-second (WPS) and mean fundamental frequency (F0).

To support downstream modeling, we also retained linguistic attributes such as transcript length and readability scores. Performance for each ASR system is shown using system-specific WER metrics, which allows detailed comparison across platforms. This consolidated dataset is used as the basis for the sentiment analysis, stance classification, and misinformation alignment pipelines described in the following methodology.

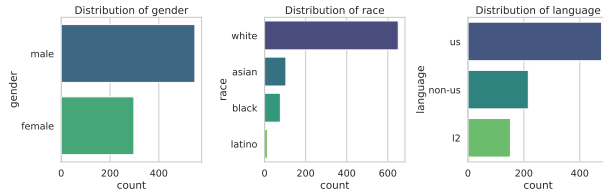


Figure 3: Gender, race, and language distribution across samples.

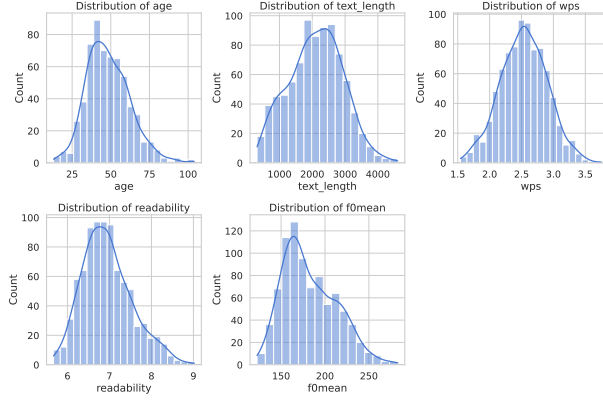


Figure 4: Distribution of numerical features across the dataset.

Fig. 4 shows the variation in linguistic and acoustic features across the 846 speakers. The differences in speaking rate and pitch vary widely across samples. These patterns do not show artificial balancing or preprocessing rather than they reflect natural variation.

Figure 3 shows distribution of demographic attributes in the final dataset. Gender distribution is imbalanced across the dataset as there are 548 samples of Male and 298 samples of Female. Racial distribution is also skewed i.e., White has the majority with 651 samples, Asian has 104 samples, Black has 77 samples and Latino has 14 samples. The Language background distribution is such that there are 478 samples of U.S. English speaking people which is the largest group among all, 216 samples of Non-U.S. English speaking people and 152 samples of L2 speakers. But imbalance is common in TED-style datasets and these disparities are important for analysis as they can impact fairness and robustness of models across demographics.

Table 3: Overview of Attributes in the Final Preprocessed Dataset.

Attribute	Description
name	Unique identifier for the speaker.
file	Filename associated with each audio-transcript pair.
transcript	Human gold-reference transcript.
text_length	Length of the reference transcript in characters.
age	Speaker age in years
gender	Gender of the speaker with values: <i>male</i> , <i>female</i> .
race	Self-identified race category with values: <i>white</i> , <i>asian</i> , <i>black</i> , <i>latino</i> .
language	English background with values: <i>US</i> (native), <i>Non-US</i> (non-native), <i>L2</i> (non-native).
wps	Words per second.
readability	Transcript complexity score computed using the New Dale-Chall formula.
f0mean	Mean fundamental frequency (Hz), estimated using MATLAB <code>pitch</code> .
YT_WER	Word Error Rate from YouTube ASR.
FB_WER	Word Error Rate from Facebook/Meta ASR.
MS_WER	Word Error Rate from Microsoft ASR.
GM_WER	Word Error Rate from Google Meet ASR.
WX_WER	Word Error Rate from WebEx ASR.
ZM_WER	Word Error Rate from Zoom ASR.
BJ_WER	Word Error Rate from BlueJeans ASR.
GM	Transcript generated by Google Meet ASR.
YT	Transcript generated by YouTube ASR.
MS	Transcript generated by Microsoft ASR.
FB	Transcript generated by Facebook/Meta ASR.
ZM	Transcript generated by Zoom ASR.
BJ	Transcript generated by BlueJeans ASR.
WX	Transcript generated by WebEx ASR.
audio_filepath	Absolute path to the audio file.

4. Methodology

This study focuses on impact of ASR transcription bias on transcripts. It examines effect on the three main downstream CSS tasks: (1) Sentiment Analysis, (2) Stance Classification and (3) Misinformation Alignment. The study follows standard ML pipeline including Data preprocessing, Feature extraction, Model training, Model evaluation and Statistical bias analysis.

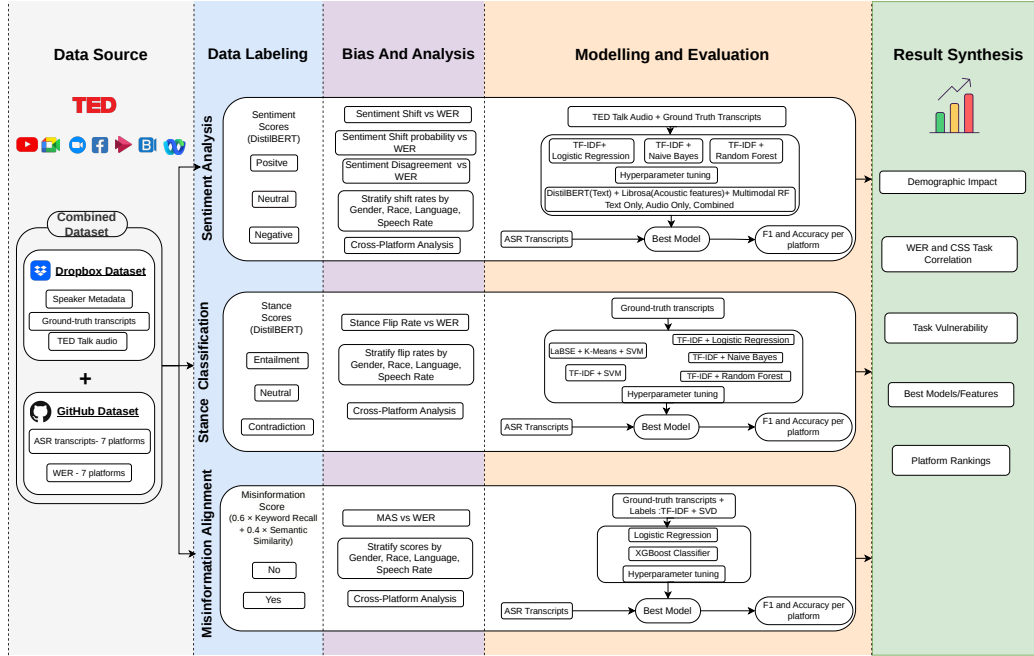


Figure 5: Overview of the Proposed ASR-to-Downstream NLP Pipeline

4.1. Sentiment Analysis Pipeline

This subsection describes how we compute gold-standard sentiment, extract ASR sentiment, build classical and multimodal classifiers, and analyse how ASR errors affect downstream sentiment robustness.

4.1.1. Gold-Standard Sentiment Score (Reference Transcripts)

We compute reference sentiment using DistilBERT (`distilbert-base-uncased`). Each transcript t produces a continuous sentiment score:

$$s_{\text{ref}}(t) = \text{SentimentHead}(\text{CLS}(t)).$$

We discretize the score into three sentiment classes:

$$c_{\text{ref}}(t) = \begin{cases} \text{positive}, & s_{\text{ref}}(t) > 0.2, \\ \text{neutral}, & |s_{\text{ref}}(t)| \leq 0.2, \\ \text{negative}, & s_{\text{ref}}(t) < -0.2. \end{cases}$$

These labels form the gold-standard reference.

4.1.2. ASR Sentiment Extraction

For each ASR system $p \in \{\text{GM, YT, MS, FB, BJ, ZM, WX}\}$, we compute sentiment using the same DistilBERT model:

$$c_p(t) \in \{\text{positive, neutral, negative}\}.$$

A **sentiment shift** occurs when the ASR label differs from gold:

$$\text{shift}_p(t) = \mathbf{1}[c_p(t) \neq c_{\text{ref}}(t)].$$

Shift rate is used to quantify downstream sentiment distortion.

4.1.3. Classical TF-IDF Models

To assess the difficulty of sentiment classification on this dataset, we train three classical machine learning models using TF-IDF features.

All classical machine learning models use TF-IDF feature extraction with `max_features = 5000`, English stop-word removal, and bigram support (*n-gram range* = (1, 2)). Hyperparameter selection is performed using 3-fold GridSearchCV with weighted F1-score.

Algorithm 1 TF-IDF Logistic Regression Training

- 1: **procedure** TRAINTFIDF_LR(T_{ref})
 - 2: Split dataset into train (80%), validation (10%), and test (10%)
 - 3: Construct TF-IDF + Logistic Regression pipeline
 - 4: Define grid: $C \in \{0.1, 1, 10\}$
 - 5: Perform 3-fold GridSearchCV using weighted F1-score
 - 6: Select best C^*
 - 7: Evaluate on validation and test sets
 - 8: **return** trained LR(C^*)
 - 9: **end procedure**
-

(a) *Logistic Regression.*

Algorithm 2 TF-IDF Multinomial Naive Bayes Training

```

1: procedure TRAINTFIDF_NB( $T_{\text{ref}}$ )
2:   Split dataset into train (80%), validation (10%), and test (10%)
3:   Construct TF-IDF + Multinomial Naive Bayes pipeline
4:   Define grid:  $\alpha \in \{0.1, 0.5, 1.0\}$ 
5:   Perform 3-fold GridSearchCV using weighted F1-score
6:   Select best  $\alpha^*$ 
7:   Evaluate on validation and test sets
8:   return trained NB( $\alpha^*$ )
9: end procedure

```

(b) *Multinomial Naive Bayes.*

Algorithm 3 TF-IDF Random Forest Training

```

1: procedure TRAINTFIDF_RF( $T_{\text{ref}}$ )
2:   Split dataset into train (80%), validation (10%), and test (10%)
3:   Construct TF-IDF + Random Forest pipeline
4:   Define grid:  $\text{max\_depth} \in \{\text{None}, 10, 20\}$ 
5:   Perform 3-fold GridSearchCV using weighted F1-score
6:   Select best depth  $d^*$ 
7:   Evaluate on validation and test sets
8:   return trained RF( $d^*$ )
9: end procedure

```

(c) *Random Forest.*

4.1.4. Acoustic Feature Extraction

Using Librosa, a 16-dimensional acoustic feature vector is extracted for each audio file. The feature set includes:

- RMS energy mean
- RMS energy standard deviation

- Pause ratio (silence proportion)
- 13 MFCC mean coefficients

These features capture loudness variation, pauses, and spectral characteristics relevant to speech prosody.

4.1.5. Text Embedding Extraction

DistilBERT generates a 768-dimensional embedding for each transcript using the first token representation:

$$\mathbf{h}(t) = \text{DistilBERT}(t)_{[:,0,:]}.$$

These embeddings serve as semantic representations for text-only and multimodal sentiment classification.

4.1.6. Multimodal Sentiment Classification

To evaluate whether acoustic information improves sentiment prediction, three Random Forest classifiers are trained:

- Text-only RF using DistilBERT embeddings
- Audio-only RF using acoustic features
- Multimodal RF using concatenated text and audio features

All Random Forest models use fixed parameters: $n_estimators = 100$ and $max_depth=10$.

Evaluation is performed using accuracy and macro-F1 on an 80–20 stratified train–test split.

4.1.7. Hyperparameter Tuning

For classical TF–IDF baselines, hyperparameters are selected using grid search with weighted F1-score.

Algorithm 4 Grid Search for Classical TF-IDF Models

```
1: procedure GRIDSEARCH( $m, X, y$ )
2:   Define model-specific hyperparameter grid  $\mathcal{H}(m)$ 
3:   Perform 3-fold cross-validation
4:   for each  $h \in \mathcal{H}(m)$  do
5:     Train model  $m(h)$ 
6:     Compute weighted F1-score
7:   end for
8:   Select  $h^* = \arg \max_h \text{F1}_{\text{weighted}}$ 
9:   return trained model  $m(h^*)$ 
10: end procedure
```

4.1.8. ASR Robustness Evaluation

To quantify transcription robustness, the best-performing reference classifier is applied to transcripts generated by each ASR platform. Performance degradation is measured relative to the ORIGINAL transcript using accuracy, macro-F1, and weighted F1.

4.1.9. Proposed End-to-End Pipeline

The complete sentiment robustness framework integrates textual, acoustic, and ASR-derived signals.

Algorithm 5 End-to-End Sentiment Robustness Pipeline

```
1: procedure SENTIMENTPIPELINE( $T_{\text{ref}}, A$ )
2:   Extract TF-IDF features from transcripts
3:   Train classical baseline models (LR, NB, RF)
4:   Extract acoustic features  $a(t)$  from audio
5:   Generate DistilBERT embeddings  $e(t)$ 
6:   Train text-only, audio-only, and multimodal RF classifiers
7:   for each ASR platform  $p$  do
8:     Obtain transcript  $t^{(p)}$ 
9:     Predict sentiment label
10:    Compute degradation relative to ORIGINAL transcript
11:  end for
12:  Analyze platform-wise robustness trends
13: end procedure
```

4.2. Stance Classification Pipeline

4.2.1. Baseline: DistilBERT MNLI Zero-shot Stance Classification

At first, we use a pretrained zero-shot classification model based on DistilBERT (`typeform/distilbert-base-uncased-mnli`).

To improve computational efficiency, predictions are generated in batches. The model is evaluated using metrics such as accuracy, F1-score, and confusion matrices.

4.2.2. Feature Representation using LaBSE

For the main pipeline, we adopt LaBSE (Language-Agnostic BERT Sentence Embeddings) to encode transcripts into numerical vectors. Each input transcript is transformed into a 768-dimensional embedding:

$$x_i = \phi(t_i) \in \mathbb{R}^{768} \quad (1)$$

where $\phi(\cdot)$ denotes the embedding function. Since LaBSE is designed for multilingual tasks, it ensures consistent representation across different languages.

4.2.3. Unsupervised Stance Discovery using K-Means

To avoid manual annotation, stance labels are generated using K-Means clustering with ($k = 3$). Each embedding is assigned to the nearest cluster centroid, effectively grouping similar transcripts together. The clusters are then interpreted as:

- Supportive
- Critical
- Neutral

These cluster assignments act as pseudo ground-truth labels for training the supervised model.

4.2.4. Supervised Stance Classification using Linear SVM

A Linear Support Vector Machine (SVM) is trained on the LaBSE embeddings to perform stance classification. The decision function is defined as:

$$f(x) = w^T x + b \quad (2)$$

The model is trained by minimizing hinge loss with regularization:

$$L(w) = \frac{1}{2}||w||^2 + C \sum_{i=1}^N \max(0, 1 - y_i f(x_i)) \quad (3)$$

where C is the regularization parameter.

The trained model is then used to classify stance in transcripts generated by different ASR systems.

4.2.5. Stance Flip Computation

To quantify semantic changes, we define a stance flip as a mismatch between predicted stance from ASR output and the original ground truth:

$$flip(i, m) = \begin{cases} 1, & \text{if predicted stance differs} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

The flip rate for an ASR system m is given by:

$$F_m = \frac{1}{N} \sum_{i=1}^N flip(i, m) \quad (5)$$

Higher flip rates indicate greater deviation in meaning caused by ASR errors.

4.2.6. Demographic Bias Analysis

We analyze whether ASR errors disproportionately impact certain demographic groups. For a group G , the flip rate is calculated as:

$$F_g = \frac{1}{|G|} \sum_{i \in G} flip(i, m) \quad (6)$$

Groups include categories such as gender (male, female), race (e.g., Black, Asian, White), and language background (native vs non-native speakers).

4.2.7. Hyperparameter Tuning

Model parameters are optimized using Grid Search with 3-fold cross-validation. The search space includes:

- $C \in \{0.1, 1, 10\}$

- Kernel $\in \{\text{linear}, \text{rbf}\}$
- $\gamma \in \{\text{scale}, \text{auto}\}$

The optimal configuration is selected based on highest weighted F1-score. Empirically, a linear kernel with $C = 1$ provides the best performance.

4.2.8. Model Validation

For robustness, stratified 5-fold cross-validation is performed. The average F1-score is computed as:

$$CV-F1 = \frac{1}{5} \sum_{k=1}^5 F1_k \quad (7)$$

We also report the mean and standard deviation of F1-scores across folds to assess consistency.

4.2.9. Pipeline

1. Collect transcripts, ASR outputs, and demographic metadata.
2. Generate baseline predictions using DistilBERT zero-shot model.
3. Extract LaBSE embeddings for all transcripts.
4. Apply K-Means clustering to generate stance labels.
5. Train a Linear SVM classifier on embeddings.
6. Predict stance for ASR-generated transcripts.
7. Compute stance flip rates for each ASR system.
8. Analyze bias across demographic groups.
9. Tune hyperparameters using Grid Search.
10. Validate performance using cross-validation.
11. Apply explainability techniques such as LIME.

4.2.10. Machine Learning Lifecycle

- **Data Collection:** Gather transcripts, ASR outputs, and metadata
- **Preprocessing:** Clean and normalize text, generate embeddings
- **Feature Engineering:** Use TF-IDF and LaBSE representations
- **Model Training:** Train models such as SVM, Logistic Regression, and DistilBERT

- **Evaluation:** Use accuracy, F1-score, and flip rate
- **Fairness Analysis:** Compare performance across demographic groups
- **Model Selection:** Choose the best-performing model
- **Explainability:** Use LIME for interpretation

4.3. Misinformation Preservation Pipeline

To Find-Out how automatic speech recognition (ASR) systems maintain misinformation-related topics, we design a pipeline by combining the semantic similarities, lexical analysis, and lightweight machine learning classifiers. This enables us to assess the accuracy with which ASR transcripts retain misinformation relevant keywords, and the semantic drift errors introduced by the generated transcriptions. This methodology follows the standard machine learning pipeline of data preparation, feature extraction, metric computation, model training, validation, and robustness analysis.

A Proposed misinformation analysis block diagram is provided below:

1. **Input:** Human reference transcripts from the audio file were paired with ASR outputs from seven different widely used platforms.
2. **Lexicon Construction:** Ranking of TF-IDF and truncated SVD projections used to extract salient misinformation-related terms and their semantic neighbors.
3. **Keyword Extraction:** In this study, reference was taken based on FlashText, and ASR transcripts were taken into consideration to identify preserved, misplaced, and hallucinated misinformation terms.
4. **Semantic Representation:** In order to capture global contextual meaning TF-IDF embeddings computed for full transcripts .
5. **Metric Computation:** Keyword Recall, Keyword Precision, Keyword F1, Cosine Semantic Similarity, Meaning Drift, and the combined Misinformation Alignment Score (MAS) derived for each transcript pair.
6. **Correlation Analysis:** Pearson correlation computed between MAS and Word Error Rate (WER).
7. **Bias and Variability Assessment:** MAS aggregated across gender, race, language background, and speech-rate categories.
8. **Chunk-Level Classification:** TF-IDF + SVD features used to train Logistic Regression and XGBoost models.

9. **Model Validation:** 80/10/10 train-validation-test split with hyperparameter tuning.
10. **Cross-ASR Comparison:** Metrics aggregated across ASR platforms.

4.3.1. Domain-Specific Lexicon Building

A misinformation-focused lexicon was constructed from the reference transcripts. TF-IDF vectorization of 1-3 grams and 250,000 maximum features was used to extract the important words and phrases. Among them, the best 6,000 terms were chosen as seeds. using Truncated SVD, Dimensionality reduction was done to get a lower semantic space rank, which helps in allowing systematic neighbor expansion. For each seed term, the top 12 semantic neighbors were retrieved using cosine similarity with a minimum threshold of 0.55 and Stop words and numerical tokens were removed. The final lexicon contains approximately 2,000 misinformation-relevant terms.

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \log\left(\frac{N}{df(t)}\right)$$

4.3.2. Keyword Matching and Metrics

Each reference and ASR transcript was processed to extract lexicon keywords. Keyword-based evaluation was computed using True Positives (TP), False Negatives (FN), and False Positives (FP).

$$\text{Recall} = \frac{TP}{TP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.3.3. Semantic Similarity Computation

Semantic preservation between reference and ASR transcripts was evaluated using cosine similarity on TF-IDF vectors. Meaning drift was defined as one minus the similarity score.

$$\text{SemanticSimilarity} = \frac{\mathbf{v_a} \cdot \mathbf{v_b}}{\|\mathbf{v_a}\| \|\mathbf{v_b}\|}$$

$$\text{MeaningDrift} = 1 - \text{SemanticSimilarity}$$

4.3.4. Misinformation Alignment Score (MAS)

To Combine lexical and semantic consistency, a Misinformation Alignment Score was defined:

$$\text{MAS} = 0.6 \times \text{KeywordRecall} + 0.4 \times \text{SemanticSimilarity}$$

The higher the MAS, the better the retention of the misinformation-related topics in ASR.

4.3.5. Similarities with Word Error Rate (WER)

By using the Pearson correlation, the relationship between Word Error Rate and Misinformation Alignment Score was obtained as:

$$r = \frac{\sum(MAS_i - \bar{MAS})(WER_i - \overline{WER})}{\sqrt{\sum(MAS_i - \bar{MAS})^2} \sqrt{\sum(WER_i - \overline{WER})^2}}$$

4.3.6. Demographic and Speech Rate Bias Analysis

To measure the group-level variation, we combine demographic fields with MAS values. Speech rate was calculated using Words Per Second (WPS) and categorised into slow, normal, fast, and very fast speech. Group-wise MAS averages help in highlighting the performance inconsistency.

4.3.7. Chunk Creation

After that, transcripts were split into 120-word chunks. If we find two or more keyword matches in the chunks, it was labeled as misinformation-rich (positive), while chunks with zero matches were labeled as negative. To reduce ambiguity, we removed chunks with a single match.

4.3.8. TF-IDF Feature Extraction

TF-IDF vectors with unigrams or bigrams having a maximum of 50,000 features and sublinear scaling were taken as the primary input feature in the process.

4.3.9. Truncated SVD (Dimensionality Reduction)

TF-IDF matrices processed through Truncated SVD in order to reduce the sparsity:

$$\mathbf{X} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$$

After reducing it, the expression becomes:

$$\mathbf{h}_i = \mathbf{U}_k(i, :)\mathbf{\Sigma}_k$$

With the help of this, we find out hidden semantics and improve the efficiency of the model.

4.3.10. Logistic Regression (LR)

Logistic Regression was chosen as a standard classifier due to its understandability and efficiency on sparse text. The model helps in predicting the likelihood of misinformation using:

$$\hat{y}_i = \sigma(\mathbf{w}^\top \mathbf{x}_i + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Hyperparameter tuning was done for regularization strength C , L2 penalty, and class weights through Grid Search.

4.3.11. XGBoost Classifier

We used XGBoost in order to find the nonlinear relationships in the TF-IDF and Reduced-SVD feature space. The model is an ensemble of boosted decision trees:

$$\hat{y}_i^{(t)} = \sum_{m=1}^t f_m(\mathbf{x}_i)$$

The training objective is:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t)$$

$$\Omega(f_t) = \gamma T + \frac{\lambda}{2} \sum_{j=1}^T w_j^2$$

4.3.12. Hyperparameter Tuning

Hyperparameter tuning was implemented by combining Grid Search (Logistic Regression) and Randomized Search (XGBoost). Table 4 explains the search space.

Table 4: Hyperparameter Search Space

Model	Parameter	Search Space
Logistic Regression	C	{0.01, 0.1, 1, 10}
	Penalty	L2
	Class Weight	balanced
XGBoost	max_depth	{3,5,7}
	learning_rate	{0.01, 0.1}
	n_estimators	{100,300,500}
	subsample	{0.6, 0.8, 1.0}
	scale_pos_weight	{1,2,5}

4.3.13. Machine Learning Life Cycle

1. **Data Preparation:** ASR and reference transcripts collected and cleaned.
2. **Feature Extraction:** TF-IDF vectors and SVD-reduced representations.
3. **Modeling:** Logistic Regression and XGBoost applied to classify misinformation-rich chunks.
4. **Training & Validation:** Train/validation/test split (80/10/10) with hyperparameter tuning.
5. **Evaluation:** Metrics include F1-score, Precision, Recall, MAS, and semantic similarity.
6. **Robustness:** Evaluation across all ASR platforms and transcript conditions.
7. **Bias Analysis:** Assess demographic and speech-rate disparities.

4.3.14. Proposed Pipeline

Algorithm 6 summarizes the proposed end-to-end misinformation preservation analysis pipeline.

Algorithm 6 End-to-End Misinformation Preservation Analysis Pipeline

```
1: procedure MISINFOPIPELINE(Transcript Set  $T$ )
2:   Build misinformation lexicon using TF-IDF ranking and SVD-based
   semantic expansion
3:   Extract keyword hits from reference transcripts
4:   Compute TF-IDF vectors for all transcripts
5:   for each ASR system  $A$  do
6:     Extract keyword hits from ASR transcripts  $T^{(A)}$ 
7:     Compute Keyword Recall, Precision, and F1 for  $A$ 
8:     Compute semantic similarity and Meaning Drift between reference
   and  $T^{(A)}$ 
9:     Compute MAS score for each transcript pair
10:  end for
11:  Compute correlation between MAS and WER for each ASR system
12:  Evaluate demographic and speech-rate variability in MAS
13:  Generate chunk-level TF-IDF + SVD features
14:  Train Logistic Regression and XGBoost classifiers on labeled chunks
15:  Perform hyperparameter tuning and validate model performance
16:  Aggregate all platform-level results for cross-ASR comparison
17: end procedure
```

4.4. Conclusion

Overall, transcription errors introduced by ASR systems consistently propagate into all three downstream tasks. If WER is high, it increases sentiment shifts, stance flips, and twisted the misinformation, especially for Latino, L2, and fast speaker categories. YT is best among all in preserving sentiment, stance, and meaning of the transcript, while WX was worst among all. Multimodal fusion improved sentiment recovery, and stance errors were most common in *CONTRADICTION* cases. Across tasks, ASR quality and demographic robustness determine how reliably meaning is preserved.

5. Results

5.1. Experiment Setup

ASR outputs from the following seven platforms were used to perform all experiments: YouTube, Microsoft, Facebook, Google Meet, BlueJeans,

Zoom, and WebEx. Train, Validation, and Test sets were based on 80-10-10 split. Grid Search or Randomized Search were used for Hyperparameter optimization where Stratified cross-validation was applied for ensuring class balance,improving model robustness,preventing overfitting and led to better generalization performance.

Table 5: Optimal Hyperparameters for All Models

Model	Optimal Hyperparameters
Logistic Regression	$C = 1$, L2 penalty, class_weight=balanced
XGBoost	max_depth=5, learning_rate=0.1, n_estimators=300
Linear SVM	$C = 1$, kernel=linear, $\gamma = \text{scale}$
Naive Bayes	default
Random Forest	n_estimators=200, max_depth=40
LaBSE + SVM	Linear SVM on 768-d embeddings
DistilBERT	Pretrained MNLI zero-shot model

5.2. Cross-Task Platform Comparison

Table 6: Comparison of ASR Platforms on WER, Sentiment Shift, Stance Accuracy, and MAS Score

Platform	WER	Shift (%)	Stance Acc	MAS	Rank
YT	0.05	10.76	0.765	0.9792	1
MS	0.061	12.41	0.7565	0.9703	2
FB	0.069	15.25	0.7116	0.9645	3
GM	0.087	15.48	0.7258	0.9597	4
BJ	0.081	15.96	0.7021	0.9637	5
ZM	0.083	15.13	0.6761	0.9536	6
WX	0.117	19.50	0.617	0.9288	7

Table 6 shows comparison of ASR platforms on key metrics. YouTube is the best among all of them as it has lowest Word Error Rate and highest Misalignment Score. Microsoft and Facebook has slightly lower performance. BlueJeans, Zoom and WebEx has higher WER. Higher WER means more sentiment shift, lower stance accuracy and weak misinformation alignment. Hence, transcription errors degrade performance of the ASR platforms.

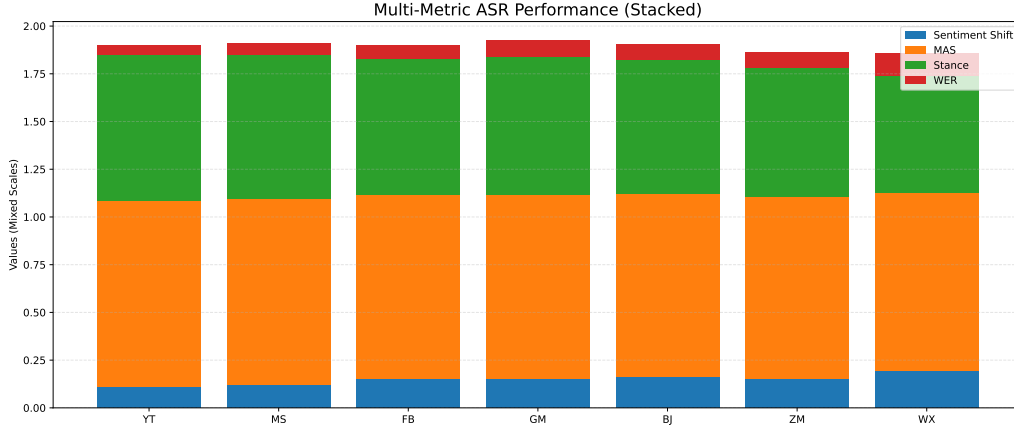


Figure 6: Multi-metric performance comparison of ASR platforms.

Figure 6 shows multi-metric comparison which are Sentimental Shift, MAS, Stance shift and WER. YouTube and Microsoft has balanced performance. Low WER causes low sentiment distortion and high stance accuracy and higher MAS. BlueJeans, Zoom, WebEx shows weaker performance. High error components have low reliability. The Clear trend is that lower WER leads to better overall performance.

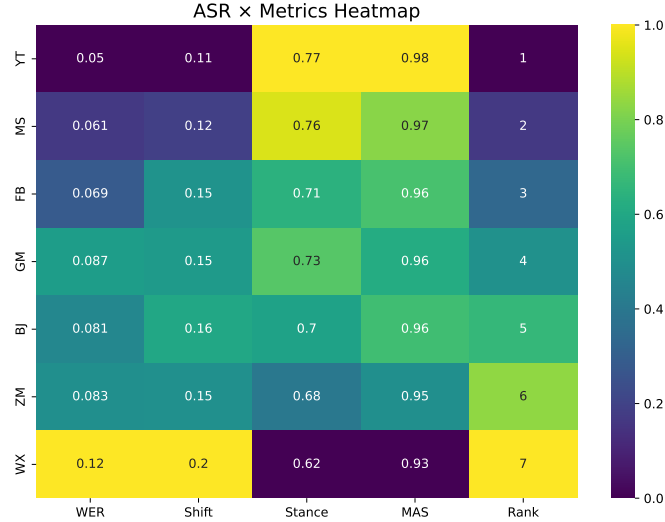


Figure 7: Platform-wise heatmap of WER, Shift, Stance Accuracy, MAS, and Rank.

Figure 7 shows Heatmap for compact multi-metric view including Metrics: WER, Sentiment Shift, Stance Shift, MAS and Rank. YouTube and Microsoft shows strong performance thus shown by cool colors. WebEx and Zoom has weaker stance accuracy hence shown by warm colors. Color gradient shows performance trend. Hence, Higher error results in poorer overall metrics.

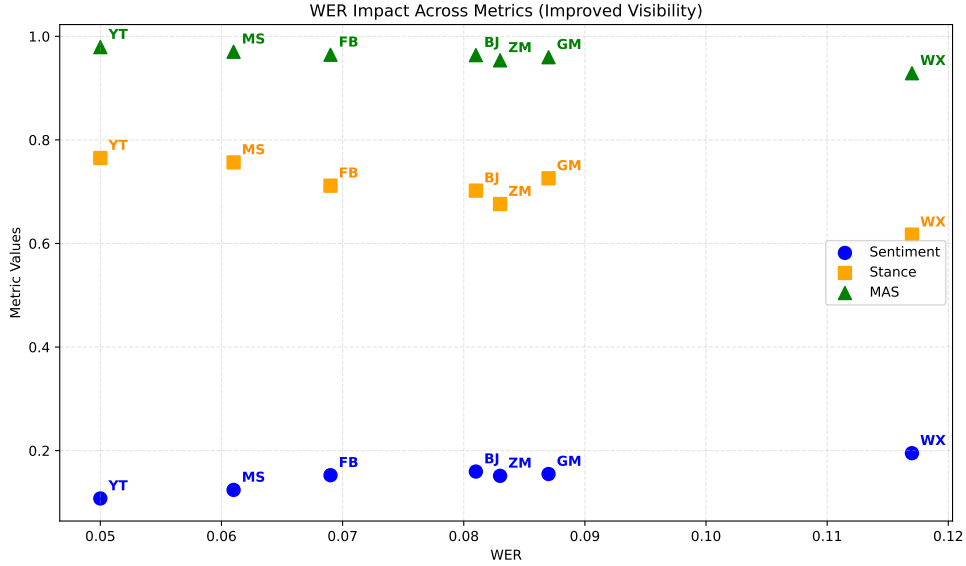


Figure 8: WER vs Sentiment Shift, Stance Accuracy, and MAS.

Figure 8 is Scatter plot for WER vs performance including Metrics: Shift, Stance and MAS. YouTube, Microsoft, Facebook shows low WER and high performance. Left cluster indicates better semantic preservation. Zoom and WebEx has high WER and shows poor performance. Small WER increase can lead to big performance drop. Therefore, transcription quality is critical.

5.3. Demographic Disparity Analysis

Table 7 shows demographic disparities including Metrics: Shift, Stance and MAS. Latino, Asian, Black, L2 speakers have high sentiment shift and MAS gap. White speakers have low sentiment shift and lower stance accuracy. Male and Female speakers shows moderate impact. Marginalized / non-native groups are more affected. Hence, ASR errors show demographic bias, impacting all downstream tasks.

Table 7: Demographic Performance and Vulnerability Across Tasks

Group	Sentiment		Stance		MAS	
	Score	Vuln.	Score	Vuln.	Score	Vuln.
Female	10.7–20.8	High	0.78	Med	0.13	Low
Male	10.8–18.8	Med	0.74	Med	0.13	Low
Latino	14.3–42.9	High	0.75–0.86	Med	1.5	High
Asian	14.4–24.0	High	0.64–0.78	High	1.3	Med
White	9.7–17.8	Med	0.69	High	0.98	Low
Black	11.7–26.0	High	0.69–0.79	Med	2.3	High
L2	11.2–20.4	High	0.611	High	2.15	High

Figure 9 shows combined demographic vulnerability index Aggregating : Shift shift, Stance degradation and MAS deviation. Race has highest vulnerability (28.6%). Gender has moderate impact (10.1%). Language has moderate impact (9.9%). ASR impact is not uniform across groups. Racial groups are affected most. Bias persists across all tasks.

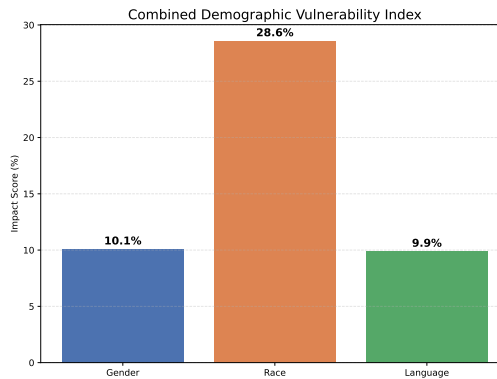


Figure 9: Demographic Vulnerability across Sentiment, Stance, and MIS tasks.

5.4. Downstream Task Reliability Analysis

5.4.1. Sentiment analysis

In this section, we examine how transcription errors from seven different platforms(GM, YT,MS,FB,BJ, ZM, WX) break the sentiment accuracy of the original TED Talk.

We first correlate the sentiment error with the WER (word error rate) to find the "tipping point", the point at which the WER impacted the sentiment shifts drastically. Our analysis does not show a correlation between emotional intensity and Word Error Rate ($r \approx 0$), indicating that ASR performance does not fail because the transcripts are emotional. Instead, we observe a distinct threshold for sentiment degradation across platforms. Figure 10 shows that this threshold emerges at $WER > 0.3$, indicating that modern ASR platforms are quite resilient to small errors. Beyond this threshold, the sentiment shift rates increase to 37.5% (WX) and 33.3% (GM).

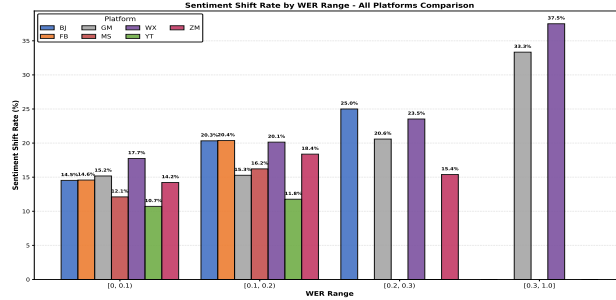
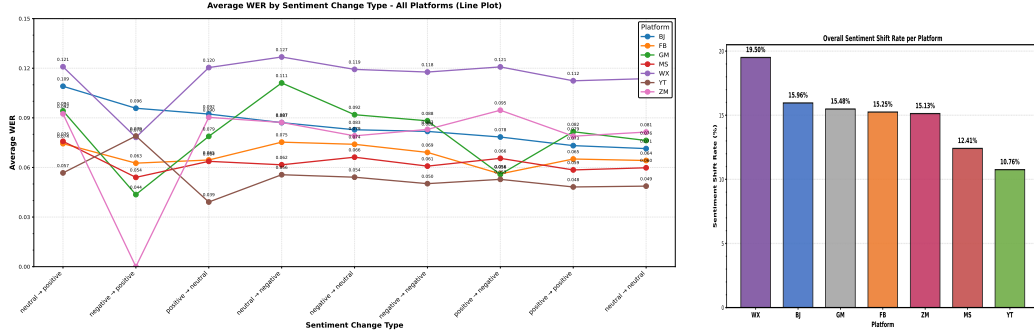


Figure 10: Threshold emerges at $WER > 0.3$ across platforms.

We further examine the type of sentiment that is most affected by WER. As illustrated in Figure 11a, the figure breaks down how these failures occur. We found that some errors affect the accuracy of the sentiment more than others. For example, the peak on the green line (GM) for the *neutral* $\rightarrow$ *negative* transition appears only at higher noise levels ($WER > 0.11$). This means that strong noise is needed to turn a neutral sentence negative. However, a *positive* sentence shifts to *neutral* at much lower error rates. This suggests that ASR errors remove a positive signal more easily than create a negative one.

Next, We analyze the impact of different groups, such as gender, race, language, and speech rate, on sentiment changes to help us understand which group is more vulnerable to sentiment changes. These failures are not distributed equally. Figure 12 highlights how ASR errors impact specific groups. We observe that the gender differences are minor. Zoom (ZM) and Webex (WX) show the most sentiment distortion when analyzing Latino and Black speakers. In addition, speakers who are not native English speakers (L2) have higher rates of sentiment shifts compared to those who are native speakers.



(a) The plot reveals the “error cost” required to trigger distinct types of sentiment inversion.

(b) Overall sentiment shift.

Figure 11: Error sensitivity across sentiment transitions and aggregate platform-level sentiment shifts.

Speech rate also affects performance. Very fast and slow speech rates lead to more sentiment errors.

In Fig. 11b we see the sentiment shift rates for platforms. We observe link between transcription quality and sentiment shifts. For example, WebEx (WX) has high error rates and shows the largest sentiment shift (19.5%). Similarly, YouTube (YT) remains the most stable with a 10.76% shift.

We then used the ground-truth transcripts and their labels. Different models are trained and evaluated, including logistic regression, Naive Bayes, random forest, and multimodal random forest. This helps identify the most robust model. We observe a severe class imbalance with 566 negative samples and only 161 Neutral and 119 Positive classes. This skew creates a critical “accuracy paradox,” as shown in Table 8. The confusion matrices (Fig. 13) provide forensic evidence of this paradox. Naive Bayes and Random Forest achieved high accuracy ($\approx 67\%$) largely by “cheating”. Driven by the imbalance stated earlier, these models simply predicted “Negative” for almost every sample (Majority Class Collapse). This resulted in a very poor Macro F1 score of ≈ 0.26 . Logistic Regression, on the other hand, actually learned to distinguish between classes. Although its raw accuracy was slightly lower (62%), its Macro F1 was significantly higher (0.417), demonstrating that it was the only model that learned the decision boundaries.

We also attempted to improve performance by adding audio features (Multimodal Fusion). However, as shown in Table 9, adding audio provided no benefit. The Audio-Only model produced a static F1 score (0.288), likely

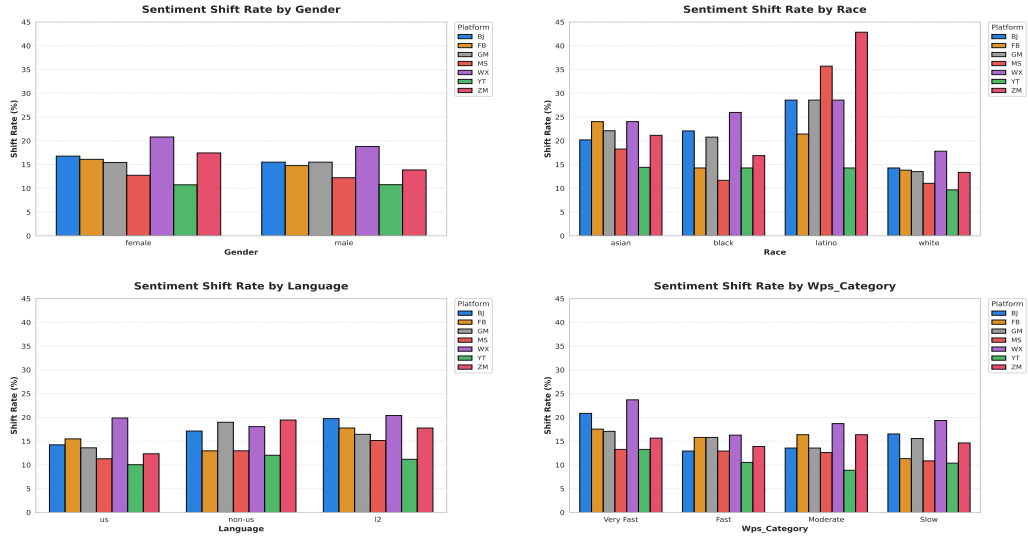


Figure 12: Sentiment shifts by groups across ASR platforms.

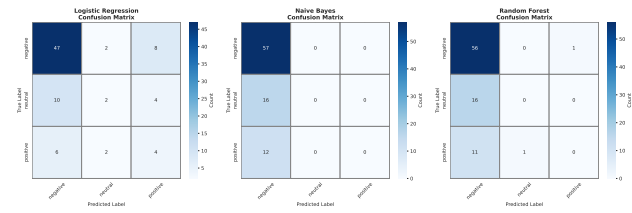


Figure 13: Confusion matrices for Logistic Regression, Naive Bayes and Random Forest.

Table 8: Performance comparison.

Model	Test Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	0.6235	0.4431	0.4276	0.4170
Naive Bayes	0.6706	0.2235	0.3333	0.2676
Random Forest	0.6588	0.2249	0.3275	0.2667

Table 9: Impact of fusion on performance metrics.

Transcript	Accuracy			Macro F1		
	Audio-Only	Combined	Text-Only	Audio-Only	Combined	Text-Only
Ground truth	0.659	0.671	0.671	0.288	0.268	0.268

acting as noise rather than signal. Because the “Combined” model failed to beat the “Text-Only” baseline, we selected **Unimodal Logistic Regression** as our final model for the rest of the analysis. Finally, we measured exactly how much ASR error hurts this optimal model. We kept the model fixed and replaced the clean Ground Truth transcripts with the noisy ASR transcripts.

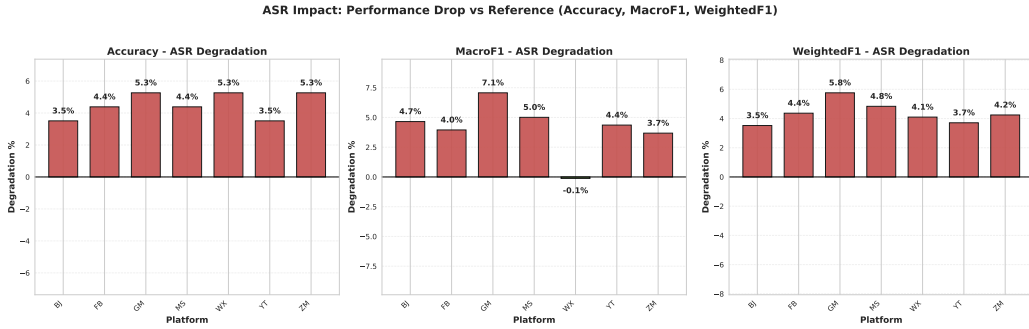


Figure 14: Relative performance degradation across ASR platforms. GM exhibits the most severe sensitivity to noise (7.1% Macro F1 drop), while accuracy uniformly degrades by 3.5–5.3% across the board.

Figure 14 shows the performance drop. Every ASR platform degraded the model’s performance. GM showed a 7.1% collapse in Macro F1, aligning with its high error rate. Table 10 confirms this trend. Platforms like BJ and YT maintained accuracy above 0.647, while the noise from platforms like GM and WX altered the accuracy.

Table 10: Performance of ASR transcripts.

Platform	Accuracy	Weighted F1
BJ	0.6471	0.6175
FB	0.6412	0.6121
GM	0.6353	0.6032
MS	0.6412	0.6091
WX	0.6353	0.6138
YT	0.6471	0.6163
ZM	0.6353	0.6129

5.4.2. Stance Classification

This sub section reports on the variation of stance prediction by race, gender, language, and ASR platforms.

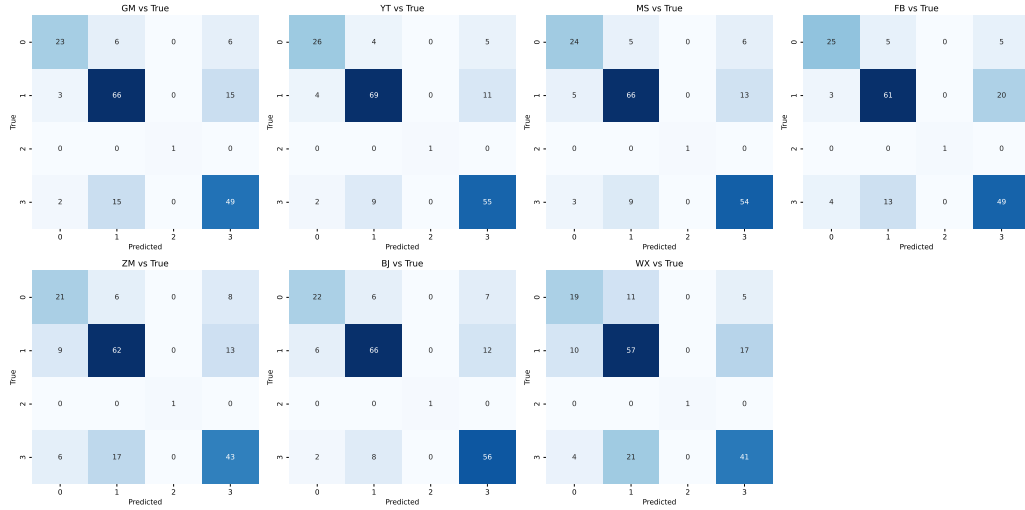


Figure 15: Classification Report. Comparison of confusion matrices across ASR models.

Figure 15 which reports that YT and MS display greater diagonal dominance which in turn indicates more consistent stance predictions. In contrast we see that WX has higher off diagonal values which reflects greater misclassification and weaker stance reliability.

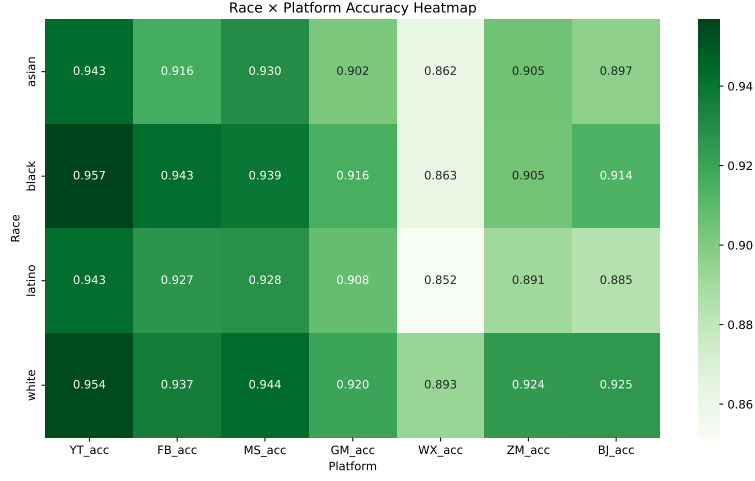


Figure 16: Race $\times$ Platform accuracy heatmap for stance classification.

The stance accuracy heatmap in Fig. 16 reports which performance we see across racial groups when we look at different ASR platforms. YouTube, Microsoft and Facebook, with values typically above 0.93, do very well across the board for all races which is to say they have great robustness in downstream stance recognition. Google Meet does a moderate job but we see it has large drops in accuracy for Asian and Latino speakers. In terms of which does the worst job overall WebEx takes the cake, in particular for Latino and Asian groups which see accuracy drop below 0.86. Zoom and BlueJeans are in the middle range, they have some moderate racial variability which still leaves them behind the best performing platforms. As a whole the heatmap shows us that we have large scale issues of race based disparity across all systems which play out in that minority groups are hit the hardest by ASR errors in stance prediction.

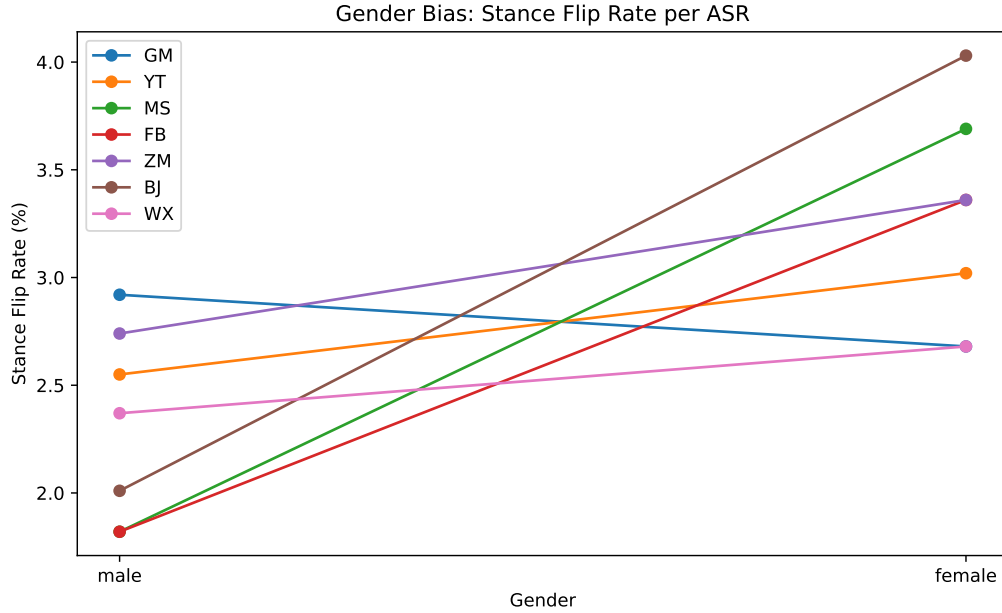


Figure 17: Gender-wise stance flip rates across all ASR platforms.

Demographic Bias and Fairness Analysis. As shown in Fig. 17, we see that stance flip rates play by gender group and ASR platform which we note to display large scale demographical bias. For the most part we see that female speaking subjects have slightly higher flip rates than that of males do better which in the case of BJ and MS' systems is very large in fact by over a percent. Also it is the case that in the instances of GM and WX which we note display the least difference in performance between genders there is also the most consistent transfer of semantics. The trend as a whole also sees that which ASR generated errors put out into the environment do not do so equally and that which systems like BJ, MS, and FB are the biggest players in terms of gender specific error generation.

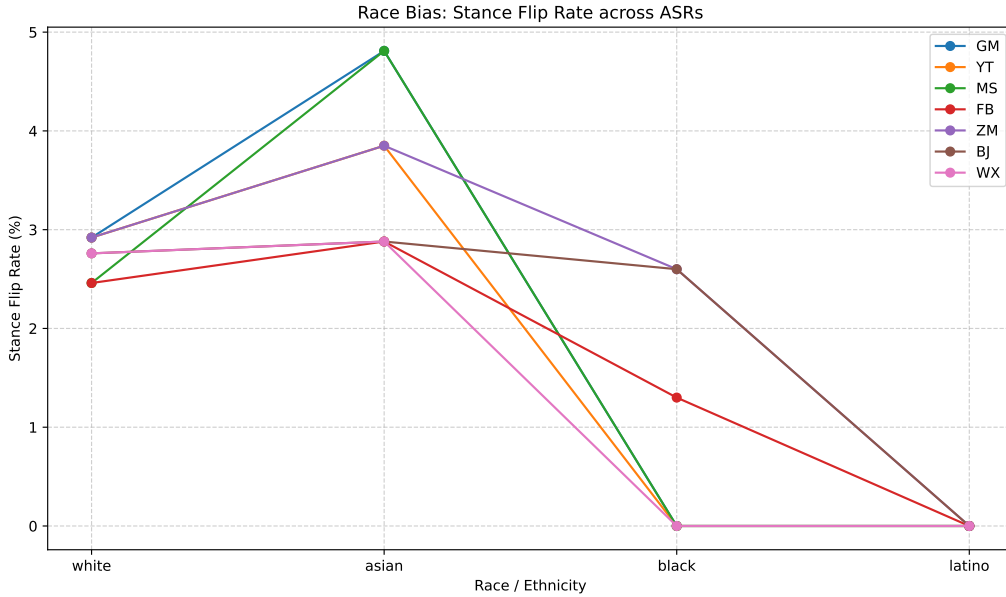


Figure 18: Race bias: stance flip rate across ASR systems.

As shown in Fig. 18, we see that stance flip rates are very different between racial groups which in turn points out that there is no uniformity in terms of ASR vulnerability. From our analysis of the race_bias data set which included most platforms for Asians we saw the highest flip rates across the board which at times hit almost 5% for GM and MS which is to say that small transcription differences play a large role in how this group is represented in terms of stance. Whites showed to have moderate flip rates in the 2.5-3% range, but Blacks and Latinos did the best with the lowest flip rates and in some cases saw almost no change at all in ASR output. Also of note is that the issue of ASR induced semantic shift does not affect all races the same way and that some demographic groups do in fact see more change in meaning based on the ASR platform used.

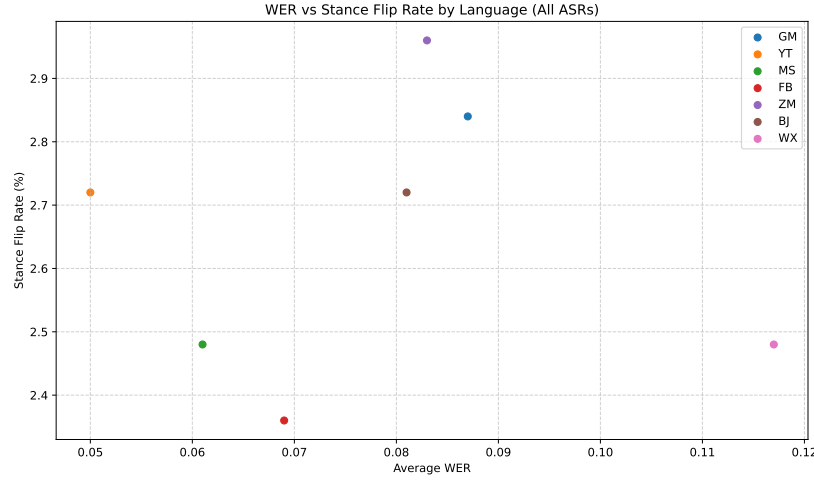


Figure 19: WER vs. stance flip rate by language category across all ASR systems.

As shown in Fig. 19, relationship strength varies by language group. US-English speakers have low WER and have lowest stance flips while non-US speakers have highest stance flips although have similar WER. L2 speakers have highest WER and stance flips. Platforms WX and ZM showed more errors for L2 group. Higher WER causes more stance distortion but it alone does not capture semantic robustness.

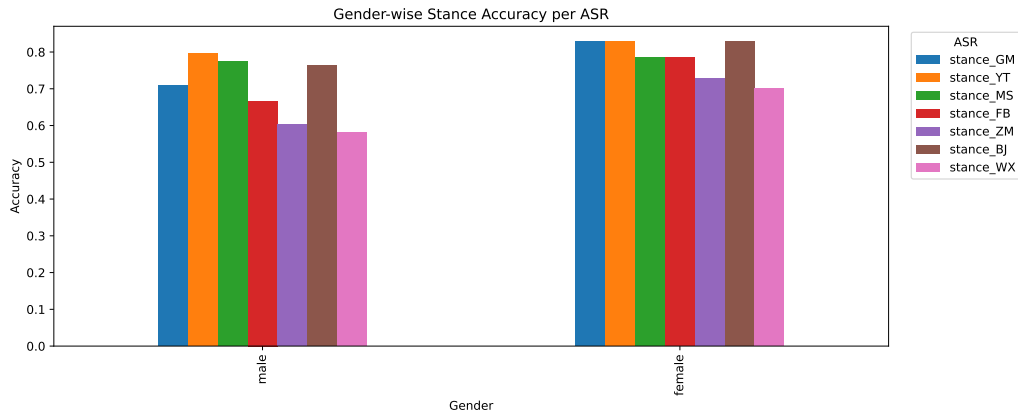


Figure 20: Gender-wise stance accuracy across ASR platforms.

The gender-wise stance accuracy patterns shown in Fig. 20 female speakers have higher stance accuracy across platforms while male speakers have

consistently lower accuracy. A gender gap is present in all ASR systems. YT, MS and BJ showed best overall performance. Female accuracy is greater than 0.82 on top platforms. WebEx had lowest accuracy overall. Accuracy for males specifically in WebEx dropped below 0.60. Zoom and Facebook had mid-range accuracy with clear gender gap.

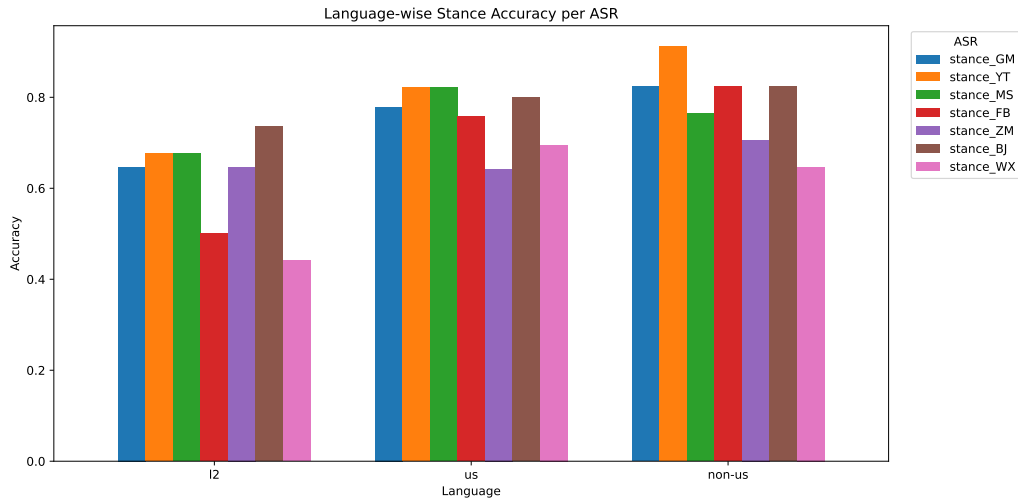


Figure 21: Language-wise stance accuracy across ASR platforms.

The language-wise stance accuracy patterns shown in Fig. 21 shows clear disparities between native and non-native speakers across all ASR systems. Accented speech is disproportionately affected. L2 speakers had lowest stance accuracy often below 0.60. Non-US native speakers had mid-range accuracy. YT, MS and BJ had greater than 0.80 accuracy specifically for US-native. WebEx and Zoom had lower accuracy for non-US speakers. Non-native speech had overall more stance errors.

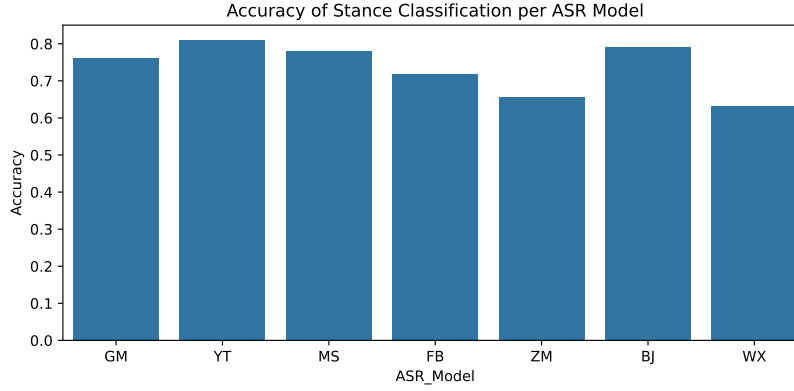


Figure 22: Overall stance accuracy per ASR platform.

The bar chart in Fig. 22 shows the overall stance classification accuracy achieved by each ASR platform. YouTube and BlueJeans show the strongest performance, with greater than 0.79 accuracy. Microsoft and Google Meet had stable mid-high accuracy between 0.76 and 0.78. Facebook and Zoom showed moderate performance. WebEx had lowest accuracy around 0.63. From all the results attained, we can conclude that better transcription leads to better stance accuracy.

5.4.3. Misinformation Alignment

We are evaluating how misinformation alignment (MAS) varies across ASR platforms and demographic groups, therefore highlighting where transcription distortions impact misinformation consistency.

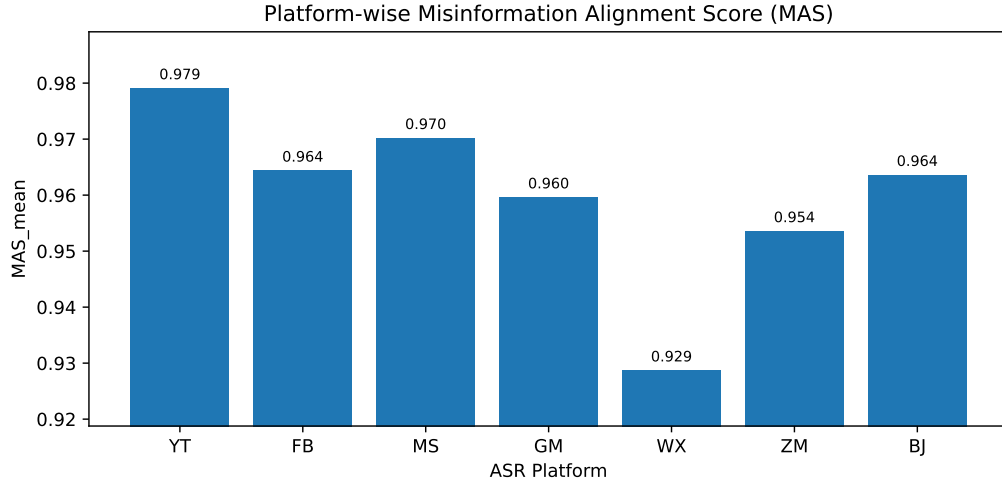


Figure 23: Platform-wise MAS scores across ASR systems.

The bar chart in Fig. 23 shows a comparison of the Misinformation Alignment Score (MAS) across all seven ASR platforms, which are YouTube (YT), Facebook (FB), Microsoft (MS), Google Meet (GM), WebEx (WX), Zoom (ZM), and BlueJeans (BJ). This chart describes how each of the said platforms maintains misinformation relevant content during transcription generation.

YouTube (YT) has the highest MAS value of 0.979, showing that its transcriptions contain the most similar semantic and keyword information alignment with the original misinformation cues. This result is associated with YouTube’s strong ASR performance and low WER, which together reduce the distortion.

Secondly, there are Microsoft (MS) and Facebook (FB), with a steady and high MAS scoring within the range of 0.964-0.970. These platforms generate a transcription that is not perfect but retains most of its basic concepts associated with misinformation.

Next, there are Google Meet (GM) and Zoom (ZM), whose MAS scores indicates that even though they capture the content correctly, they are prone to losing the actual meaning, leading to incorrect information. This reflects their moderate WER levels.

Blue Jeans (BJ) is slightly lower than the previous category. Here, stability is quite good, though still lagging behind the better ones. It is not as prone to error as WebEx, though BJ struggles to maintain fine details in

misinforming.

The least stable model at this end and with the lowest MAS score of 0.929 is WebEx (WX), which needs to be mentioned first. The presence of this significantly higher tendency towards distorting its misinformation-sensitive parts, mainly as a result of its relatively high WER, is owing to missing keywords, incorrect substitutions, and semantic changes.

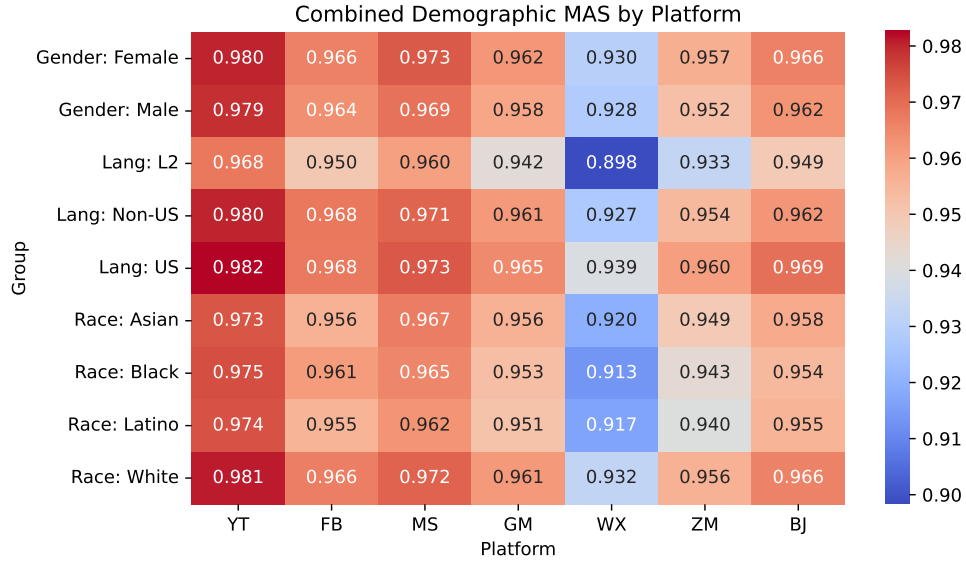


Figure 24: Combined demographic MAS heatmap.

Figure 24 summarizes how misinformation alignment (MAS) differs across demographic groups for each and every ASR platform. YouTube (YT), Microsoft (MS), and Facebook (FB) perform consistently high MAS scores across gender, race, and language categories, showing reliable preservation of misinformation alignment among all the speaker variations. In contrast, WebEx (WX) demonstrates the largest demographic changes, having notable drops for L2 speakers and a few racial groups, suggesting higher susceptibility to content distortion.

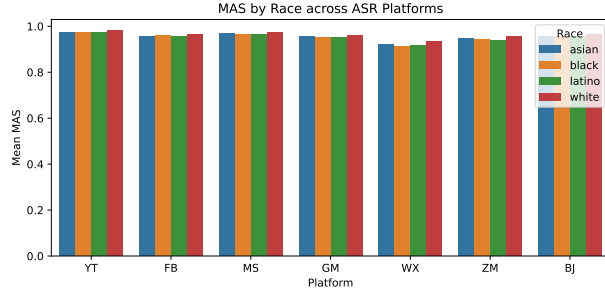


Figure 25: MAS disparities across race groups.

According to Fig. 25, the misinformation alignment remains regularly high across all racial groups, with the dependent variations being low among platforms. YouTube, Microsoft, and BlueJeans show the highest stability, consistently maintaining MAS values above 0.97 for most of the groups. In contrast, WebEx consistently shows lower alignment across all races, confirming its weaker robustness.

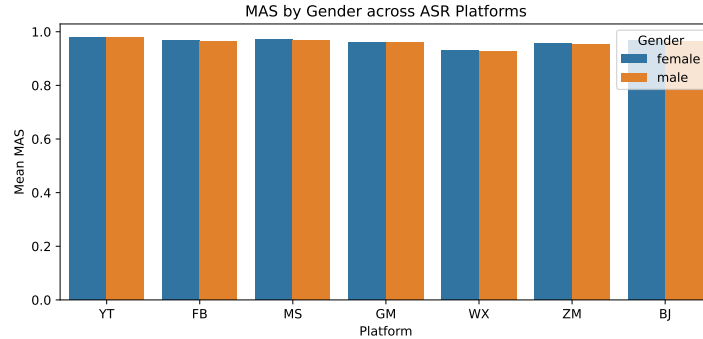


Figure 26: MAS disparities by gender across platforms.

As shown in Figure 26, for both male and female speakers, the misinformation alignment remains consistently high across all ASR platforms, showcasing low gender-based variation (typically within 0.5–1%), revealing that MAS is largely insensitive to gender-specific speaker or linguistic characteristics. Performance-based differences are primarily driven by platform-level transcription quality rather than speaker gender.

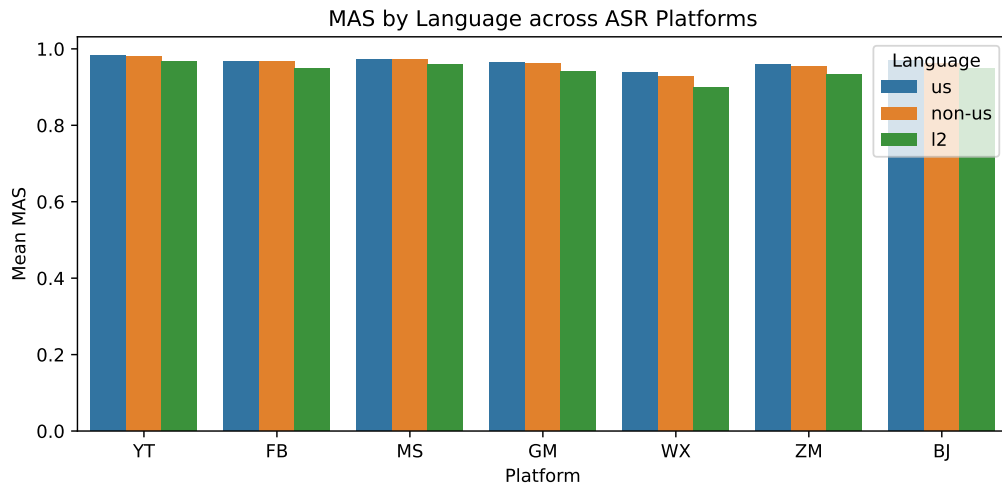


Figure 27: MAS differences across language groups.

The language wise MAS comparison in figure 27 shows that MAS is consistently high across all the language groups which are US, non-US, and L2, indicates a moderate impact of linguistic background on the stability of MAS. US and non-US speakers have generally high MAS scores above 0.96 on most platforms, whereas L2 users have relatively lower alignment, typically 0.93–0.96, especially on weak systems like WebEx and Zoom. Stronger ASR platforms such as YouTube, Microsoft, and BlueJeans have uniformly high MAS performance across all the languages, whereas platforms with higher WER have more noticeable drops for L2 users. Overall, these findings suggest that MAS is influenced by language proficiency, but rather slightly than stronger demographic factors of variation, such as race.

As shown in Fig. 28, the change in Logistic Regression (LR) recall between gold and ASR transcripts is extremely small, typically on the order of 10^{-4} . Among the platforms, WebEx and Facebook exhibit the largest degradation, while Microsoft shows almost no loss, indicating strong classifier robustness to ASR-induced noise.

Table 11 summarizes keyword retention behavior across ASR platforms. While many transcripts achieve perfect keyword preservation, the lower-bound F1-score of approximately 0.78 indicates that a subset of transcriptions suffers substantial lexical loss, directly impacting misinformation alignment quality.

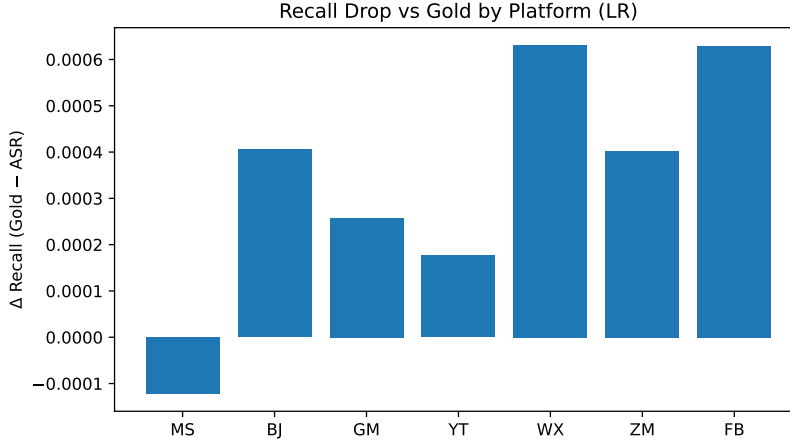


Figure 28: LR recall drop between gold and ASR transcripts across platforms.

Table 11: Range of keyword-matching performance across ASR platforms.

Metric	Min	Max	Gold Reference
Keyword recall	0.678	1.000	1.000
Keyword precision	0.690	1.000	1.000
Keyword F1-score (approx.)	0.780	1.000	1.000

5.5. Model Selection

The model comparison in Table 12 shows performance differences between base and high approaches across sentiment analysis, stance classification, and misinformation alignment tasks. Normal approaches such as Random Forest and Naive Bayes show moderate performance, but Logistic Regression provides strong results, with highest MAS scores for misinformation alignment. For stance classification, the LaBSE + SVM model delivers the best accuracy (0.952 ± 0.015 0.952 ± 0.015), outperforming all base models used for text analysis. XGBoost also performs better on the MIS task, but stance capability of the model shows limitations. Random Forest used for the integration of the audio and text features, performs less on sentiment. Overall, advanced embedding based models especially LaBSE+SVM and Logistic Regression are the most reliable one with the best results and for the scalability purposes

Table 12: Model Comparison Summary Across All Tasks

Model	Sentiment	Stance	MIS	Features	Scale
Random Forest	0.659/0.537	0.647/0.610	N/A	TF-IDF	Med
Naive Bayes	0.671/0.538	0.659/0.633	N/A	TF-IDF	High
Logistic Regression	0.624/0.600	0.753/0.748	0.9994/0.9997	TF-IDF	Very High
SVM (TF-IDF)	N/A	0.765/0.762	N/A	TF-IDF	Med
LaBSE+SVM	N/A	0.952 $\pm$ 0.015	N/A	LaBSE	Med
XGBoost	N/A	N/A	0.9903/0.9951	TF-IDF	Med
Multimodal RF	0.658/0.291	N/A	N/A	MFCC+TF-IDF	Low

The results in Table 12 and Fig. 29 show the differences across the classical, baselines and multimodel used for the evaluation of the sentiment, stance and MIS classification. For sentiment analysis, classical TF-IDF models show strong class imbalance results in the moderate accuracy. Random Forest and Naive Bayes generate majority class biases and Logistic Regression(LR) shows the most stable trade offs (0.624 ACC, 0.600 F1), resulting in the most scalable and reliable text baseline.

For **sentiment analysis**, due to the strong imbalance in the dataset, the TF-IDF models did not perform well. Random Forest and Naive Bayes models show high and majority class bias with low macro F1 despite moderate accuracy. Logistic Regression(LR) shows the most stable trade offs (**0.624 ACC**, **0.600 F1**), resulting in the most scalable and reliable text baseline. The Random Forest performs inefficiently (**0.291 F1**) because many features add noise, resulting in decrease in the effective and useful feature ratio in the dataset. For sentiment prediction on ASR generated transcripts, the Logistic Regression model performs the best among all the classical text models.

For **stance classification**, TF-IDF models provide normally good separability (0.762–0.765 F1). However the advanced embedding models are the ones that perform the best. The LaBSE + SVM model shows the best overall stance performance (**0.952 $\pm$ 0.015 F1**), which remains robust across the different possible featuring hurdles like ASR noise and demographic variation. Logistic Regression ranks second (0.748 F1), however Random Forest and Naive Bayes performs less table and scalable due lexical features.

misinformation alignment, depends majorily on keyword matching. Again, Logistic Regression performs best and perfect scores (**0.9994 ACC**, **0.9997 F1**). XGBoost performs competitively (0.9951 F1) but shows greater

sensitivity to demographic shifts, especially for non-native (L2) speakers. Multimodal RF combining MFCC + TF-IDF underperforms across tasks due to acoustic variability and the dominance of textual cues in misinformation detection.

Overall, the best models offer the strongest performance, highest cross-platform robustness, and minimal demographic disparities, as summarized visually in Fig. 29.

Table 13 summarizes the final hyperparameter settings and feature extraction techniques for the best-performing architectures across all three downstream tasks.

Table 13: Optimal hyperparameter configuration and feature extraction settings for the best-performing models.

Downstream Task	Component	Configuration & Hyperparameters
Sentiment Analysis	Feature Extraction	TfidfVectorizer: Max Features=5000, N-gram=(1,2), Stop Words='english'
	Model (LogReg)	$C = 1$, Solver='lbfgs', Class Weight='balanced', Max Iter=200, Random State=42
Stance Classification	Clustering	KMeans: $k = 3$ (Support, Neutral, Oppose), Random State=42
	Classification	SVM (Linear): $C = 1$, Gamma='scale', Kernel='linear'
Misinformation Alignment	Feature Extraction	TfidfVectorizer: Max Features=50,000, N-gram=(1,2), Sublinear TF=True, Lowercase=True
	Model (LogReg)	$C = 100.0$, Solver='lbfgs', Penalty='l2', Class Weight='balanced'

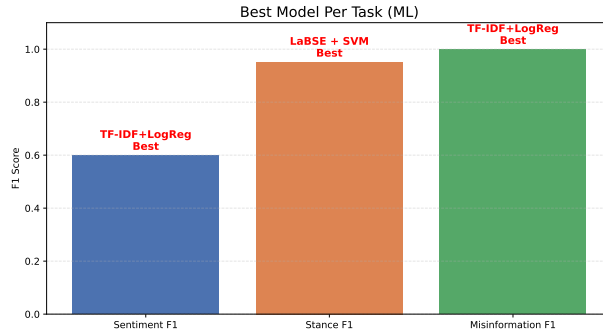


Figure 29: Model comparison across sentiment, stance, and misinformation tasks.

5.6. Final Multi-Metric Summary

As shown in Fig. 30 on radar visualization, it gives the comparison of all ASR platforms on five key metrics such as WER, sentiment shift, stance accuracy, MAS, and overall rank weight. The model clearly depicts that the **YouTube (YT)** consistently achieves the Strong position among all the dimensions, with **lowest WER**, **minimal sentiment distortion**, and **highest stance and MAS performance**. **Microsoft (MS)** is the next most balanced platform, having high stance and MAS scores and moderate shift and WER levels. **Facebook (FB)**, **Google Meet (GM)**, and **BlueJeans (BJ)** depict mid-range performance, having clear drops in stance and MAS compared to YT and MS but without extreme differences. In contrast, **WebEx (WX)** forms the smallest radial changes, reflecting its **highest WER**, **largest sentiment shift**, and **lowest stance and MAS stability**, being overall the weakest platform. This radar chart highlights the strengths and weaknesses of each ASR system in a single multidimensional view.

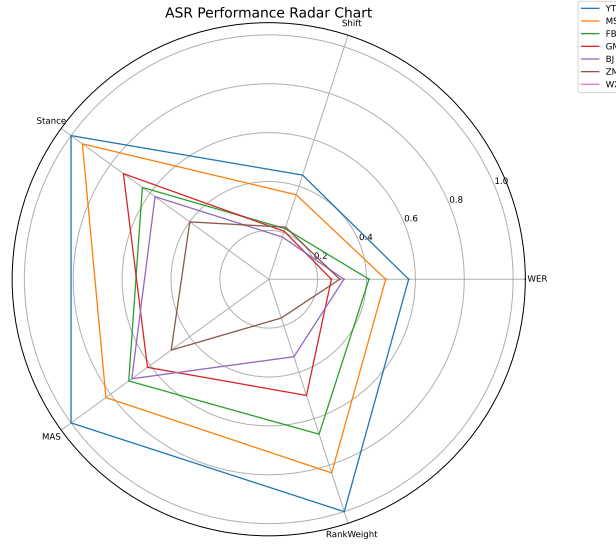


Figure 30: Radar Plot: Multi-Metric Platform Profile.

6. Discussion

6.1. Key Findings

Automated transcription errors extend beyond lexical inaccuracies. They affect sentiment, stance, and misinformation alignment. Platforms with lower WER maintain the original intent. While, platforms with higher WER, such as WebEx and Zoom, often alter the intended meaning of the speaker. Although modern platforms are resilient to small errors but after a certain WER threshold the original meaning changes drastically. These findings suggest that WER should not be treated as a standalone metric, it is a predictor of reliability as well.

Further we see that distortions are not evenly distributed. Speakers such as Latino, Asian, Black, and L2 speakers consistently experience greater semantic drift. In this as well, we observe that groups with higher WER showed more semantic drift than those with less WER. These findings indicate that ASR errors are not random and shape downstream interpretive outcomes.

The stance and misinformation findings show how a small change like an incorrect negation or omission of a word in automated transcripts lead to great shifts in meaning. The dependency of downstream metrics on semantic accuracy is shown by the behaviour of the Misinformation Alignment Scores. MAS relies heavily on semantic consistency and the keyword matching, so whenever there is some keyword mismatching or content dissimilarity the MAS scores decline. At low WER levels there is drift from the Positive to neutral, whereas high error rates shows the transition from neutral to negative.

These findings conclude the key insights from ASR and NLP research are robustness requires diverse training data, meaning sensitivity which shows the importance of semantic preservation and the demographic difficulties remain a huge challenge.

6.2. Comparison With Related Work

Table 14 shows our results and highlights how our findings takes prior work on ASR fairness one step ahead, and downstream NLP robustness and scalability.

Table 14: Comparison With Related Studies

Study	Focus Area	Key Finding	Relation to Our Work
Cao (2023) [1]	ASR robustness in noisy settings	Across speakers and environments, ASR systems vary widely in accuracy	Our Stance and MAS robustness analysis extend this work into semantic calculation.
Mujtaba (2024) [7]	ASR fairness and demographic bias	Significant accuracy disparities across accents and demographic groups	We observe similar demographic disparities, but measured at the semantic/stance level.
Wu (2022) [11]	ASR errors and downstream NLP	Minor ASR errors cause sentiment/semantic misclassification	We confirm this effect for stance and misinformation detection.
Dubois (2024) [2]	Demographic WER bias at scale	Large, systematic demographic differences in ASR error patterns	Our work extends beyond WER to show how these differences propagate into stance and MAS distortions.
This Work	ASR + stance + misinformation	Semantic preservation varies by system and speaker	Integrates platform-level, semantic-level, and demographic comparisons not jointly examined before.

6.3. Broader Implications

Implications for ASR Research. Traditional metrics like WER underestimate the real impact of transcription errors. Our findings demonstrate that semantic preservation—not merely surface accuracy—is essential for evaluating ASR systems, especially in politically or socially sensitive domains.

Implications for Platforms and Policymakers. The different demographics are revealing some concerns with the variability of semantics and how it pertains to the fairness of the analysis of the content moderation systems, the systems flagging misinformation, and the automated political system analyses.

Implications for NLP Pipelines. The concerns are coming from how the ASR systems are utilized, and with the use of ASR systems, research bias can be created in the analysis of NLP pipelines. ASR transcripts can be used with some limitations, especially with the downstream processes. The transcription errors, especially if they are on the smaller side, can be detrimental to the analysis of the stances, and, therefore, a decision can be made in the real world that may be detrimental, based on the error. The transcription errors can be on the smaller side and still be detrimental. The people doing research on the analysis of stances may become frustrated on the basis of how the errors are acting.

6.4. Limitations

Several issues and what in large part shape the results. For one, we see that ASR induced bias is a great issue. Some ASR systems report to have a semantic drift for certain demographics which in turn may be to put certain groups at a greater disadvantage in what comes after.

Secondly, we see that which elements make up the data set is a issue we are dealing with we have a large gap between the different class populations which in turn plays into how the models perform. We see mainly large groups represented which in term causes the models to over fit to those and at the same time have very little information to work with regarding the smaller groups. While the overall accuracy may look good what we see in Macro F1 scores and in the confusion matrices is a large degree of the model’s performance is on account of it failing to properly deal with the minority classes which in the end does not give us a robust or generalizable model for stance and sentiment.

Third, the zero-shot classifier is sensitive to noisy transcripts. Hence, minor wording changes leads to stance reversal. There exists an embedding-based inference instability.

Additionally, there is a domain mismatch as pretrained models are not optimized for political or misinformation speech. Therefore, reliability is reduced on nuanced ideological content.

Moreover, there exists a limited language and accent coverage. Accent or dialect diversity is restricted. Cross-linguistic generalization is limited.

Also, WER has insufficiency as similar WER has different stance outcomes. Surface accuracy is not equivalent to semantic preservation. WER cannot explain semantic drift individually.

6.5. Future Work

Future research should explore several directions such that robustness and fairness evaluation can be improved. This could be expanded to more languages and dialects. Spontaneous conversational speech can be included with a broader demographic representation.

Addressing the issue of class imbalance in downstream stance classification is still very much a priority. In terms of future work we should see stronger rebalancing strategies like focal loss, class aware sampling, and synthetic minority augmentation which in turn will prevent minority class collapse. We put forth that improving class separability would in fact also improve stance stability and fairness which in the case of imperfect ASR results is very true.

Also in this space we see promise in the use of prosodic information. Elements like pitch, stress, and intonation which often present key stance clues that which do not get full play in our text transcriptions. Also we should look into Multi modal audio text frameworks which combine acoustic features in to the mix as they may reduce semantic drift and improve interpretive fidelity.

Also at present we see that end-to-end task aware ASR systems which are joint optimization for the downstream stance issues could fix up some of the defects by having transcription output align with what is relevant in terms of semantic structure. Also we should see the development of fairness oriented semantic evaluation metrics which in turn improve or replace the Word Error Rate. Metrics which put forth a comparative analysis of how different demographic groups interpret what is said will go beyond the Word

Error Rate in its present form and in that which they put into practice will present a more relevant measure of the real world impact.

7. Conclusion

This work provides comprehensive analysis of stance classification and sentiments, examines misinformation alignment affects and focus on semantic preservation. It is found that transcription quality directly affects meaning preservation. Errors influence downstream stance outcomes and effects vary across ASR platforms. Meanwhile, it is found that demographic groups experience unequal impact.

It can be reported that high-performing ASR shows strong semantic robustness whereas weak ASR shows semantic robustness. The study is important for public discourse applications. It is relevant to moderation systems and impacts civic speech analysis. Moreover, it affects automated decision-making. As ASR becomes increasingly integrated into social and informational ecosystems, it should be ensured that transcription systems preserve speaker's intent meanwhile fairness, transparency and trust is maintained as they depend on semantic fidelity.

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